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**A Project Report
on**

AyurSage

**Submitted for fulfillment of project work
of**

BACHELOR OF TECHNOLOGY

Degree in

COMPUTER SCIENCE AND ENGINEERING

By

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UNDERTAKING

1 The work presented in our dissertation titled “**AyurSage**”, submitted to the Department of Computer Science and Engineering, Hindustan College of Science and Technology, Mathura, for the award of Bachelor of Technology, is our original work. We have neither plagiarized nor submitted the same work for award of any other degree. In case this undertaking is found incorrect, we accept that our degree may be withdrawn.

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Certified that ADITI VERMA, DIKSHA SHARMA, JAGVEER SINGH BEDI, YASHI UPADHYAY have carried out their Project work presented in this Report entitled “AYURSAGE” for the SEVENTH SEMESTER PROJECT EVALUATION of Bachelor of Technology from Dr. A.P.J Abdul kalam Technical University, Lucknow under my supervision. The Report embodies result of original work and studies carried out by Student himself and the contents of the Report do not form the basis for the award of any other degree to the candidate or to anybody else.

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ABSTRACT

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14
Ayurveda, an age-old holistic medical science explains health as dynamic equilibrium of three Doshas viz. Vata, Pitta and Kapha. Although the practice is very prevalent, it is not yet integrated with modern digital healthcare due to the lack of structured datasets and computational tools that can interpret classical Ayurvedic knowledge. This project, AyurSage: An AI-Based Ayurvedic Consultation System addresses this gap by using the technique of Text extraction from Ayurvedic textbooks to build a structured dataset for symptom–Dosha mapping. These structured mappings, along with the contextual features like age, stress, diet, sleep and seasonal factors form the basis of training a Random Forest classifier to predict the Dosha imbalances using the user inputs.

16
The implementation is done using a full stack architecture, in which a Flask backend is used to host the machine learning model and connect to a React-based frontend user interface that will be used to collect symptoms and provide customized Ayurvedic recommendations. AyurSage not only automates Dosha prediction, but also forms real guidance on herbs, diet and lifestyle changes with the foundation in Ayurvedic principles. This project is one way of demonstrating the feasibility of digitizing Ayurvedic diagnostics using the AI-driven methodologies and provides a framework for making computational Ayurveda scalable. The work is paving the way for improvements in the future, namely multi-dosha classification, advanced disease mapping, and integration with health devices for wearables, and it is part of the evolution of personalized digital wellness.

Keywords: Ayurveda, Dosha Prediction, Text Extraction, Machine Learning, Symptom Extraction, Digital Health, Personalized Recommendation System.

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CHAPTER – 1

INTRODUCTION

36 Ayurveda, which is made up of the Sanskrit terms "Ayur" (life) and "Veda" (knowledge), is amongst the oldest holistic healing systems in the world. It offers an all-round perspective on health by focusing on the health of the body, mind and emotions through herbal medicines, life style and individualized treatments depending on body constitution (Prakriti) [1]. Over a period of thousands of years Ayurveda has been proved in cases of chronic disorders, lifestyle diseases, immunity building and preventive healthcare.

28 However, despite its immense potential, Ayurvedic diagnosis remains largely expert based, traditional manual based and scattered information based. For a common user to find proper Ayurvedic guidance is very difficult as the content on the Internet are not consistent, nor are they organized scientifically and there is no single unified digital platform.

AyurSage strives to overcome these limitations and combines the latest machine learning techniques with real Ayurvedic knowledge. AyurSage is built as an AI-powered system that understands the symptoms provided by a user and predicts the potential diseases along with the underlying dosha imbalances (Vata, Pitta, Kapha) and accordingly recommends Ayurvedic medicines and lifestyle and diet recommendations. The system performs the role of a virtual Ayurvedic assistant and offers reliable, data-based predictions in a matter of seconds [2].

In the modern world with the rise of digital technology, people have been increasingly looking towards natural medicine and preventive healthcare. This trend gives rise to a big demand for a structured platform that offers genuine Ayurvedic recommendations with data intelligence [3]. The driving force behind the project is to make Ayurveda easily accessible, understandable and tailored to each person regardless of geographic and economic barriers.

26 AyurSage is a web application implemented as a full stack using React.js as the frontend with Node.js and Express.js as the backend and a machine learning inference server implemented in Python using Flask. The platform uses two autonomous ML models, the first one to predict the most prevalent Dosha imbalance (Vata, Pitta, or Kapha) using reported symptoms and lifestyle factors of the user and environmental factors, and the second one to determine the most probable disease according to the predicted Dosha. The models are trained over filtered Ayurvedic data and provide confidence-scored predictions so that only good and trustworthy ones are given to the end user. The

system also translates these forecasts into a formal recommendation framework which proposes diet, herbal treatments and lifestyle changes in line with the classical Ayurvedic books.

The users are able to evaluate their permanent body-mind type by participating in a 30 question Prakriti Analysis module, review their evaluation history on a personal dashboard, and to make appointments with certified Ayurvedic practitioners using an integrated consultation platform. The platform would be secured using JWT based authentication, where all data of users, and health records will be kept confidential and the authenticated user can only access the information. AyurSage is a worthwhile move in the right direction to democratize preventative care and make personalized Ayurvedic advice accessible to anyone and everywhere by integrating old Ayurvedic knowledge with new software engineering and artificial intelligence.

1.1 Introduction of the Origin of Ayurveda and Digital Health

Ayurveda is one of the most ancient systems of medicine that offers holistic understanding towards the human health through an emphasis on balance among the physical, mental, emotional and environmental factors of life. Originating over five millennia ago, Ayurveda is based on the very basic concept that every individual has a unique constitutional makeup called Prakriti which is formed by the **three basic Doshas (Vata, Pitta and Kapha)** [4]. These Doshas which are responsible for the bodily functions like movement, metabolism and structural stability of the body have to be kept at a state of equilibrium for the optimal healthy condition of the body. Any disturbance in their natural balance generates Vikriti, or imbalance, which develops itself in symptoms and ends up in the appearance of disease. Classical texts like Charaka Samhita, Sushruta Samhita describe in detail the frameworks of observing these imbalances by keen observation of symptoms, lifestyle, digestion habits, emotional tendencies and the environmental responses [5].

In today's era, there are now various technologies in the area of digital health that have helped make the landscape of healthcare different with applied artificial intelligence, predictive analytics, remote monitoring systems, and mobile medical applications. These innovations have made it possible to have fast diagnostics, virtual consultations and personalized treatment paths. However, Ayurveda is yet to become a part of this digital change because of unstructured nature of its knowledge and the practitioner dependent interpretation [6]. Traditional diagnosis of Ayurveda relies on the judgment of the doctor based on the subjective characteristics and hence it is not easy to design a computational

model. The absence of structured data and a defined way of mapping a diagnosis complicates further the efforts for the digitization of the insights of Ayurveda. AyurSage solves this issue by introducing an AI-based consultation system that combines ancient Ayurvedic knowledge with the prediction power of machine learning and the knowledge extraction power of Retrieval-Augmented Generation (RAG). The integration with digital systems symbolizes not only a technological innovation, but also a large step towards getting traditional healthcare accessible in modern forms.

The Ayurveda and digital health convergence is a rare chance to restore and distribute traditional medicine to the global audience. Although data-driven methods in modern medicine have proved extremely helpful, the existing traditional systems such as Ayurveda have been mostly left behind by computational researchers because the diagnostic methods adopted by traditional systems are qualitative and highly individualistic. The recent developments in natural language processing, supervised learning, and structured medical ontologies have provided new opportunities in translation of Ayurvedic principles into machine-readable formats. With translations of classical symptom-Dosha relations into structured data, predictive models can be trained that are similar to the diagnostic thinking of a skilled Ayurvedic practitioner, which means that the scope of this ancient science can be extended to modern technology. Moreover, the increasing use of preventive healthcare patterns in the world is in line with the Ayurveda philosophy and the view of its primary benefit, which is to maintain balance and not treat the disease once it occurs. Governments and health organizations in most countries around the globe are quickly accepting the importance of incorporating traditional system of medicine into the modern healthcare systems. **The Ministry of AYUSH in India has** been at **the** forefront to encourage digital programs to enter Ayurveda, Yoga, Unani, Siddha and Homeopathy into the digital age. AyurSage is a part of this larger trend by providing an evidence-based platform that is technologically sound and has a significant gap between the classical Ayurvedic concepts and the modern digital health infrastructure, allowing personalized wellness advice to be accessible to users no matter their location or economic status.

1.2 Need for AyurSage

The growing interest in the world around natural healing systems and personalized health solutions has made there be a high demand for easy-to-reach Ayurvedic consultation. However, people tend to struggle with the ability to seek genuine Ayurvedic advice because of limitations such as practitioners, geographic locations, and financial restrictions [7]. Even with the practitioners there may be variations in Ayurvedic diagnosis between the practitioners due to the interpretive nature of

Ayurveda. This inconsistency leads to confusion for the users who want to get clear and personalised insights on his or her symptoms and his or her Dosha balance.

The digital arena does have some internet sites that do offer general Ayurvedic information, but very few, if any, offer personalized diagnostic support based on the symptoms and lifestyle patterns of the user. Most applications currently exist provide static information instead of individualized assessment. There is also a fundamental technological challenge involved - Ayurveda doesn't have machine-readable datasets mapping the symptoms to Doshas or Ayurvedic diseases in a structured format that can be used in artificial intelligence systems [8]. This absence prevents us from developing accurate machine learning models that can be used to make Ayurvedic diagnosis. AyurSage is responding to this need by creating a dataset by AI-driven extraction of Ayurvedic knowledge and employ this dataset in a predictive model. It offers an organized, automated and scalable platform to deliver personalized insights in Ayurveda, democratizing access to holistic wellness.

The AyurSage is even more warranted through a public health context of the growing burden of lifestyle diseases like obesity, diabetes, hypertension, and anxiety disorders, which Ayurveda has a long and rich history of preventive and management guidelines. Traditional healthcare models can be described as largely reactive and focused on treating problems once they are clinically relevant. Ayurveda, in its turn, is based on the preventative scheme according to which the imbalances are recognized and fixed before it transforms into a disease. An online application that would help to diagnose the current condition of a person in terms of Dosha and give him or her the appropriate recommendations at a certain time would be an early intervention tool that would help decrease the cost of relying on medical care and help him or her maintain his or her health over time. Also, the AyurSage requirement is relevant to the sphere of health literacy and patient empowerment. The large part of the world population does not know about their own special body constitution and cannot make a wise choice regarding nutrition, sleep, physical activity, and coping with stress. AyurSage helps people to gain an insight into their own health in a purposeful and practical sense by giving its users an organized Prakriti evaluation and the interpretation of their Dosha type. This process of passive health consumption to active health participation is a decisive consequence of the platform, which is guaranteed so that people are not just consumers of instructions, but informed participants in their wellness process.

1.3 Problem Statement

Ayurveda provides a complex and extremely insightful system for the diagnosis and treatment of human health conditions by considering the imbalances of Doshas, lifestyle pattern, environmental interaction, and manifestation of symptoms. However, despite its holistic and holistic personalized approach, Ayurveda has not been integrated into modern technological systems whole-heartedly due to lack of structured datasets or uniform diagnostic tools and computational methodologies that can interpret the classical knowledge effectively. Traditional Ayurveda diagnosis requires a lot of expert practitioners that test a patient through a multi-dimensional approach by examining the symptom pattern, digestive levels, emotional disposition, and lifestyle conditions. This method, although effective, is subjective, not scalable, and not accessible to a large portion of the population due to geographical and economic limitations.

The main issue solved by AyurSage is a lack of an intelligent system, which can understand symptoms provided by users, in a way that is true to Ayurvedic principles and provide accurate personalized Ayurvedic recommendations to users. There is no standardised digital data set that correlates symptoms to dosha imbalances and there are also no computational models that can process classical Ayurvedic knowledge at scale. Existing platforms have general information, but fail to provide meaningful diagnostic support. Therefore, the challenge lies in building a full-fledged system that can gather knowledge from traditional ayurvedic texts and organize it in dataset, train accurate machine learning models and display the results with the help of an intelligent and interactive interface. AyurSage wants to fill this gap by building a single system through which data extraction, Dosha prediction, and personalized recommendation generation are integrated by relying on true Ayurvedic theory.

Another aspect of the issue is that there is no centralized site that can bridge symptom-focused evaluation and practical health-related advice and professional care under one seamless process. The existing digital applications address them as distinct issues, as certain apps provide symptom checkers, others have informational resources on Ayurveda, and some provide the opportunity to book an appointment. None of them combine all three functions in a logically, data-driven system., it is not an easy task to construct a dataset that fully represents these dimensions and at the same time is consistent with classical Ayurvedic theory.

1.4 Objectives Of the Project

The development of AyurSage needs clarity and objectives in just what is achieved from transforming the concepts of traditional Ayurveda to models which can be accessed computationally.

The main goal is to develop and propose an Ayurvedic consultation system to extract symptoms, predict the dosha imbalance and recommend customized Ayurvedic remedies depending on structured input data. To achieve this goal, a number of interrelated objectives have been set out in this project, which taken together aim to realise the overarching vision of the digitization of Ayurvedic diagnosis and its accessibility to a global audience.

Objective 1: Data Extraction & Structuring

A basic goal of this project is to gain knowledge from the real-time textual sources of ayurveda and to make them into a structured data set that can be processed by machine learning methods. Since no publicly available dataset is available in Ayurveda, the project retrieves the relationship of symptoms and doshas from classical texts, research articles, and validated online Ayurvedic repositories. The extracted content has to be converted to a standard format that is helpful to maintain consistency, accuracy and ease of computation. This organized dataset is the backbone of the machine learning model, and is crucial to the attainment of the level of diagnosis accuracy. Developing a reliable dataset helps to keep AyurSage true to classical rules of Ayurveda and not to modern interpretations or subjective presumptions.

Objective 2: Developing The Machine Learning Model

Building a strong and valid machine learning model is crucial in order to predict Dosha imbalances given input structured features. This objective training and testing a strong machine learning models that can make predictions on the dominant dosha as well as the most likely Ayurvedic disease category. Random Forest and XGBoost ensemble techniques were used to guarantee high predictive power and retain explainability which is a critical attribute of clinical decision support systems.

Objective 3: Real Time Ayurvedic Consultation Interface

A final objective of the project is to create a user-friendly, interactive, and dynamic consultation interface, where the users can input their symptoms, view predictions and get personalized Ayurvedic recommendations. This interface, built with React.js, must provide a good way to navigate, have multi-step forms, allow for symptoms to be entered into the system, and even have animations for the output to provide an engaging look for the users. Flask based backend will connect the trained models and the MongoDB database to provide real time dosha and disease predictions and their respective treatment and lifestyle recommendations. The real-time linking of

the frontend interface to the backend prediction system helps in ensuring that the users get instantaneous results and actionable insights.

1.5 Project deliverables

AyurSage is designed to offer a holistic set of output that cuts across the area of data extraction, machine learning, system integration and user experience. A major deliverable is a formal and validated Ayurvedic data with the Ayurvedic symptoms mappings to Doshas based on classical Ayurveda. This dataset can be used both academically and technologically as it helps to conduct research work in the area of computational Ayurveda in the future and helps to develop the predictive models. The other significant deliverable would be a fully trained machine learning model capable of predicting the Dosha imbalance with a fairly stable accuracy. This model is implemented in the back-end system and is applied in the production of real-time predictions depending on user inputs. Besides that, AyurSage offers a fully operational Web-based consultation system with React.js as the client-side framework and Flask as the machine-learning server with a layer of Node.js and Express.js API. This platform will be comprised of the following modules: a symptom entry module, a prediction interface, an individualized recommendations section and a user dashboard to review previous performance. One of the final deliverables is a complete technical documentation and project report on system architecture and techniques, implementation and evaluation findings.

In addition to the fundamental technical deliverables, AyurSage also creates a series of practical user-facing products that serve as evidence of the usefulness of the system. These consist of a 30 item Prakriti assessment module that identifies the permanent body-mind constitution of a user, a multi-step Dosha prediction form that gathers the lifestyle, dietary, and symptomatic data and a result report that displays the predicted Dosha imbalance and the most probable associated disease and Ayurvedic recommendations. All these modules are discrete deliverables that could be assessed individually on the criteria of accuracy, usability, and clinical relevance. On the quality assurance aspect, the platform will have input validation on both frontend and backend levels in the system to make sure that only meaningful and complete data is entered to the machine learning models. The system would also include the use of user authentication via JWT, using secure API between the front and backend server and using MongoDB Atlas to have persistent data storage. A final deliverable is a full technical documentation and project report providing system architecture and methodologies, implementation details and evaluation results. These deliverables restore over all

of AyurSage as a solution that is scalable, accessible and academically rigorous contribution to digital Ayurveda.

1.6 Importance of the proposed system

AyurSage is relevant because it has the potential to transform the access to Ayurvedic diagnosis, its accuracy, and personalization using artificial intelligence. Through its project of transforming the unstructured knowledge of Ayurveda into a structured computational dataset, AyurSage provides a potential base of the future research and technological progress within the domain of digital Ayurveda. The system brings a methodological change in that the diagnostic reasoning, which is presented in the books of Ayurveda, is systematically coded into machine-readable characteristics such that the predictions are valid, transparent, and repeatable. This will be used as precedent on how modern technologies can be used to understand and scale more traditional medical systems, which could not be previously modelled using more traditional computational techniques.

In addition to that, AyurSage will address an urgent need in the society since there will be people without access to qualified Ayurvedic therapists or in areas where Ayurveda is not commonly practiced. The system allows users to know their Dosha imbalance and lifestyle triggers and assume preventive healthcare measures through classical Ayurvedic principles. AyurSage is also part of the larger project of creating integrative medicine as a new approach to comprehensive health that is the combination of the merits of both the traditional and data-driven systems by combining both diagnostic theory and Ayurveda recommendations with the modern machine learning.

At a bigger level, AyurSage is an academic contribution to the expanding body of research of customized health systems. Its methodological framework, which is a combination of structured data engineering, supervised machine learning and user-centric interface design, can be applicable to other traditional medical systems or other health practices worldwide. The system is thus not only important as a stand-alone application but also as a model that can be duplicated by researchers, technologists and healthcare policy makers who want to take the holistic healthcare practice into the digital age.

The significance of AyurSage is also supported by the fact that it may influence the preventative health outcomes on a population level. The rising health problems in the world, especially Type 2 diabetes, hypertension, stress-related disorders, and digestive problems are classified as lifestyle diseases. Ayurveda has written down guidelines on how to treat and prevent all these diseases by use of individualized diet, daily routine, herbal supplements and mind-body measures. AyurSage can become a scalable preventive health tool that can replace traditional medical care and decrease the

overall load on the healthcare systems by making those protocols available using a digital interface that personalizes the recommendations based on the symptom profiles.

1.7 Why Symptom Extraction is Important to the Diagnosis

The Ayurvedic diagnosis is based on symptoms extraction and interpretation because in Ayurveda, health imbalances are never perceived as the existence or lack of a single symptom. All the symptoms reflect more profound physiological change that is connected with Dosha behavior. To illustrate, the roughness, dryness, insomnia, or anxiety are typically indicators of a Vata imbalance; the signs of heat, irritability, or excessive thirst are indicative of Pitta unbalance; the heavy digestion, lethargy or congestion are typical of Kapha imbalance. Ayurveda is founded on the holistic knowledge of these patterns as opposed to it being an individually occurring signs [9].

It is not an easy task to extract the symptoms of classical Ayurvedic texts since they are narrative and descriptive. The terms used in the ancient sources are metaphorical and situational and it is hard to translate them directly into machine-readable format. The process of symptom extraction becomes critical to close the gap between the classical Ayurvedic knowledge and modern computerized techniques. Unless symptoms are extracted and structured correctly, a machine learning model would not be able to reflect the actual logic of Ayurvedic diagnosis and the predictions made based on this logic would not hold much value [10].

AyurSage solves this issue by selectively refining a systematic database where user-reported symptoms and lifestyle parameters are systematically cross-linked to their respective Dosha categories and risk factors of diseases, so that the interpretation can still be in agreement with classic Ayurvedic knowledge. AyurSage does not just rely on simple keyword matching to do the process of symptom extraction. The system gathers multi-dimensional data of user inputs that encompass physical traits, eating habits, sleep patterns, stress management, seasonal reactions and subjective descriptions of symptoms. Every one of these input dimensions is associated with certain diagnostic parameters identified in Ayurvedic texts that enable the system to create a complete profile of the current health status of the user. Such multi-parameter model resembles the holistic evaluation that an Ayurvedic practitioner carries out and makes sure that the machine learning model is fed with rich enough input features to make valid and clinically relevant prediction.

Moreover, it is the quality of the symptom information that will directly define the reliability and clinical usability of the predictions that AyurSage produces. To cope with the variability in symptoms which the user should report, structured input forms which are filled with predefined choices of major parameters are used in the platform to include sleep quality, stress level, diet type

and seasonal sensitivity. Such a method of data collection is controlled and thereby ambiguity is minimized as well as consistency among various users and sessions. Simultaneously, an open-text field of symptom description enables users to add more details that the model can take into consideration in order to narrow down the assessment. Such a synthesis of structured and descriptive contributions reflects the two-sidedness of Ayurvedic diagnosis systematized and descriptive at the same time - which is at the core of the AyurSage platform accuracy and reliability.

1.8 Approach to Data Extraction in AyurSage

The AyurSage data extraction process follows a discipline-oriented approach, which focuses on the authenticity of text and clinical uniformity. It has a systematic methodology, which involved disease-dosha-symptom pattern identification in three Ayurvedic canonical texts. These books were scanned and broken down to extract pertinent verses that talked about the pathogenesis, symptomatology, and treatment options. All the extracted records were recorded referring to the source of the text, making them transparent and traceable. The resulting data was then standardised to structured format that contained sixteen attributes, which fell into contextual and clinical domains. Controlled augmentation methods were used to boost the diversity of patient profiles without the need to reduce the semantic accuracy. This method produced a quality dataset with the capability of training explainable ensemble-based machine learning. The codified information is now a sound basis of further computational discovery of Ayurvedic concepts, as it allows the computerization and analytical investigation of ancient medical knowledge. [11].

Creating a reliable dataset for an Ayurvedic system requires a structured approach because no ready-made database meets the detailed format needed. To achieve this, information is gathered directly from classical Ayurvedic sources like the Charaka Samhita and Sushruta Samhita. The workflow starts by extracting usable text from chosen books. Next, it identifies key elements such as symptoms, doshas, herbs, and treatments using pattern matching and basic natural language processing. Once these elements are extracted, they are organized into a framework with sixteen fields covering patient profiles, diagnoses, and treatment recommendations. After this initial conversion, the entries go through a validation process to check for accuracy, consistency, and relevance. This step involves reviewing terminology, removing duplicates, standardizing herb names, and ensuring that the recommendations follow Ayurvedic principles [12].

The entire process takes about four weeks and uses a mix of automated techniques and careful manual review. By the end, the goal is to create over 1,000 detailed, high-quality cases suitable for AI model training. Each entry links back to its original Ayurvedic source, ensuring transparency and

authenticity. This disciplined approach results in a clean, organized dataset that is research-ready for predictive modelling tasks such as diagnosing diseases and suggesting treatments based on symptoms. This systematic strategy not only supports the current project but can also be expanded for future development [13].

ID	Age	Gender	Prakriti	Symptoms	Stress Level	Sleep Pattern	Diet Type	Season	Climate	Dosha	Disease	Therapy	Medicines	Diet Plan	Exercise
1	21	Female	Pitta	constipation, bloating, insomnia	High	Interrupted	Mixed	Varsha	Hot	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
2	22	Female	Vata-Kapha	anxiety, dry skin, bloating	Low	Normal	Mixed	Varsha	Dry	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
3	26	Male	Pitta-Kapha	sluggishness, lethargy, weight gain	Moderate	Interrupted	Non-vegetarian	Shishir	Coastal	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
4	36	Female	Vata-Kapha	constipation, bloating, dry skin	Moderate	Interrupted	Mixed	Hemant	Hot	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
5	24	Male	Pitta	bloating, insomnia, anxiety	Moderate	Insomnia	Non-vegetarian	Shishir	Coastal	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
6	42	Female	Vata	anger, acne, inflammation	Moderate	Normal	Vegetarian	Varsha	Hot	Pitta	Skin Erupti Shirodhara, Virecha	Neem, Am Bitter gree	Cooling prana		
7	57	Female	Pitta-Kapha	bloating, insomnia, anxiety	Low	Hypersomnia	Vegetarian	Shishir	Humid	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
8	52	Male	Vata	insomnia, bloating, dry skin	Moderate	Normal	Mixed	Hemant	Coastal	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
9	40	Male	Vata-Kapha	acne, acid reflux, anger	High	Normal	Vegetarian	Varsha	Dry	Pitta	Skin Erupti Shirodhara, Virecha	Neem, Am Bitter gree	Cooling prana		
10	18	Female	Vata-Kapha	red eyes, acne, inflammation	Moderate	Hypersomnia	Non-vegetarian	Vasant	Hot	Pitta	Skin Erupti Shirodhara, Virecha	Neem, Am Bitter gree	Cooling prana		
11	22	Female	Pitta	congestion, lethargy, weight gain	Low	Hypersomnia	Mixed	Shishir	Hot	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
12	38	Male	Vata-Kapha	mucus, weight gain, lethargy	Moderate	Hypersomnia	Mixed	Shishir	Hot	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
13	36	Female	Pitta-Kapha	anxiety, constipation, insomnia	Moderate	Interrupted	Non-vegetarian	Shishir	Dry	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
14	18	Female	Kapha	sluggishness, weight gain, mucus	High	Interrupted	Non-vegetarian	Sharad	Cold	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
15	56	Female	Vata-Kapha	red eyes, acid reflux, anger	Moderate	Normal	Vegetarian	Hemant	Dry	Pitta	Skin Erupti Shirodhara, Virecha	Neem, Am Bitter gree	Cooling prana		
16	56	Male	Vata-Pitta	red eyes, anger, acne	High	Hypersomnia	Mixed	Varsha	Coastal	Pitta	Skin Erupti Shirodhara, Virecha	Neem, Am Bitter gree	Cooling prana		
17	29	Male	Pitta	sluggishness, weight gain, lethargy	Moderate	Insomnia	Mixed	Vasant	Dry	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
18	36	Female	Vata-Kapha	weight gain, mucus, congestion	Moderate	Insomnia	Vegetarian	Sharad	Dry	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
19	49	Male	Vata	sluggishness, lethargy, congestion	High	Normal	Mixed	Sharad	Coastal	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
20	28	Female	Vata-Kapha	congestion, lethargy, sluggishness	High	Insomnia	Non-vegetarian	Grishma	Hot	Kapha	Hypothyroidism	Nas Guggulu, T Light, dry f Surya Namaskar	Triphala, A Warm, mo	Gentle yoga, r	
21	49	Male	Pitta-Kapha	anxiety, insomnia, constipation	High	Normal	Vegetarian	Hemant	Hot	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
22	56	Male	Pitta-Kapha	red eyes, anger, acid reflux	Moderate	Hypersomnia	Mixed	Varsha	Dry	Pitta	Skin Erupti Shirodhara, Virecha	Neem, Am Bitter gree	Cooling prana		
23	51	Female	Vata-Kapha	dry skin, insomnia, constipation	High	Normal	Vegetarian	Shishir	Hot	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
24	28	Female	Vata	insomnia, anxiety, bloating	Moderate	Interrupted	Non-vegetarian	Sharad	Coastal	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
25	49	Female	Kapha	dry skin, bloating, insomnia	High	Normal	Non-vegetarian	Sharad	Mountain	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	
26	19	Male	Kapha	insomnia, bloating, anxiety	High	Normal	Mixed	Grishma	Coastal	Vata	IBS	Basti, Abhyanga	Triphala, A Warm, mo	Gentle yoga, r	

Figure 1: Dataset

1.9 Scope of the Project

AyurSage will embrace the entire diagnosis of Ayurvedic consultation starting with the gathering of the symptoms and ending with Dosha forecasting with customized Ayurvedic prescriptions. The project consists of the sophisticated machine learning models of Dosha classification and disease prediction, and the modern React-based user-friendly interface of interactive consultations. AyurSage aims at providing users a personal perspective of their own Dosha imbalance and giving real advice on herbs, eating habits, lifestyle modification and preventive care activities. It is supposed to be employed as an initial consultation device to shift the direction towards holistic well-being as opposed to a replacement of qualified Ayurvedic practitioners [14].

The modular architecture is simple to add more sophisticated disease prediction features, a broader dataset, conversational health assistants, or even add mobile apps to the application in the future. AyurSage has a technical scope of a complete web application that is designed on a three-tier architecture. The frontend will be built with React.js, which offers a dynamic and interactive user interface wherein customers can take their Prakriti assessment, provide their inputs on Dosha analysis and see personalized health report. The API layer is developed in Node.js and Express and it handles user authentication, data storage in MongoDB Atlas and interpreting the communication between the front and the machine learning server. The machine learning inference layer is deployed

as a standalone Flask server in Python and it hosts the trained classification models and makes available prediction endpoints which are invoked by the backend in real time. Such a separation of concerns makes it possible to maintain, test and scale each of the layers of the system separately. It is also an area of the project that cuts across to the data engineering aspect of the project. An Ayurvedic database has been formulated to correlate the symptoms and lifestyle variables and environmental variables that are reported by users to Dosha types and risks of diseases. This data is the basis of **training and evaluation of the machine learning models. The** procedure of **the** data collection and structuring was done in a meticulous correspondence to the classical Ayurvedic models such that the mappings are medically exact and in accordance to the principle of traditional diagnostics. This is not only the data that is utilized in AyurSage, but the dataset can also serve as a useful academic resource to future researchers in the context of computational Ayurveda.

The scope, in terms of the user experience is comprised of the design and development of four major functional modules: a Prakriti Analysis module to assess constitutional, a Dosha Prediction module to detect real-time imbalance, an Expert Consultation module to book a session with certified Ayurvedic practitioners and a personal Dashboard to view the assessment history and health progress over time. The modules are created so that each one is usable and accessible, and all users without any prior knowledge about Ayurveda can use the platform without issues. The scope does not extend to include emergency medical advice, clinical diagnosis of severe pathological states, or prescription of pharmaceutical interventions, as it has a strict division between the functions of the platform as a wellness guidance tool and the field of licensed medical practice.

1.10 Organization of the Report

The report has seven chapters that are systematised to give a comprehensive view of the conceptual basis of AyurSage, its methodological rigour and its implementation. The first chapter gives a background of the philosophical and technological interests behind the project, which forms the setting under which AyurSage has been developed. The second chapter is a review of the existing digital applications in health sector, Ayurvedic apps and scholarly research work where the gaps that justify the need of having such a system as AyurSage are presented. The third chapter will consist of the problem statement, goals and importance of the study. Chapter four gives the system design such as architecture diagrams and workflow models. The fifth chapter describes the methodology applied to create the datasets, preprocess the data, and train the **machine learning** model. **In the** sixth chapter, **the** author **describes the** implementation specifics **of** the system, such as the frontend development, the design of the backend API, and the implementation of the model. The seventh chapter gives the

findings and analysis of AyurSage. The last chapter is the summary of the report and the direction of the future development is outlined [15].

The chapters are organized in such a way that they continue on the topics of the preceding ones to make sure that the reader has a progressive and comprehensive view of the project, starting with the theoretical basis through to the practical conclusions of the project. Chapter 2 literature review provides the scholarly and technical background of the work and uncovers the research gaps that AyurSage addresses and places the project in the context of the larger picture of AI-assisted healthcare and digital traditional medicine. Chapter 3 then converts these contextual results into an accurate problem statement, the actual technical and societal problems that the system itself is created to address and specifies the objective measurability of success. Chapter 4 and Chapter 5 are the design and methodology chapters, which are the technical heart of the report. Chapter 4 includes the high-level system architecture, the data flow diagrams and the module interaction models, which specify the way the different components of AyurSage will be structured and interrelated.

Chapter 5 gives a comprehensive description of the process of constructing the Ayurvedic dataset, selecting and engineering input features, and training and validating and optimizing the machine learning models. The combination of these chapters offers adequate technicalities of the work to be copied or expanded upon by subsequent researchers. Chapters 6 and 7 then illustrate how the design and methodology was put into practice and give the full implemented system and assess the performance based on the predetermined accuracy, usability and functional criteria.

The final chapter is the synthesis of the research results of the whole project and the determination of the tangible prospects of the further extension of the scope and abilities of AyurSage in the further work.



CHAPTER – 2

LITERATURE SURVEY

The purpose of the literature survey is to analyze the existing research, digital healthcare tools and ayurvedic application to understand the limitation and find opportunity for innovativeness. This chapter discusses the existing technological scenario of Ayurveda-oriented platforms, analyzes the capabilities of these platforms, and identifies the collected research gaps that provide the justification for the need of AyurSage- an AI powered Ayurvedic disease prediction and recommendation system.

2.1 Presentation of Existing Systems

Digital healthcare has been through a huge evolution this past decade with the explosion of mobile health applications, artificial intelligence-based diagnostic tools, and telemedicine platforms. However, despite this pervasive digital revolution, Ayurveda - a system of medicine that is based on centuries of traditional knowledge of medicine - has not been equally reflective of this technological revolution. Existing Ayurvedic systems available in the internet today are substantially different from one another in terms of their purpose, complexity, and depth. Some of these platforms merely offer textual information while others try to offer preliminary assessments or product recommendations [16]. Nevertheless, none of these systems incorporate these concepts of artificial intelligence and structured data sets, as well as personalised diagnostic modelling in accordance with classical Ayurvedic diagnostic principles, in their totality. Understanding the landscape of systems that are in place is important to put the contribution of AyurSage to the larger context of digital ayurveda.

2.2 Static Ayurvedic Information Websites

The oldest digital representation of Ayurveda was in the form of static informational websites. These platforms describe information concerning general description of the doshas, prevalent diseases, herbal methods, diets, and seasonal norms. Even though there been useful resources for learning, they are not personalized nor they're diagnostic. Users are given generic content regardless of their symptoms, lifestyle patterns or Prakriti. These websites tend to have overly simplified versions of Ayurvedic principle, which has the potential to cause an understanding and misuse of the remedy. In addition, static platforms lack mechanisms for extracting the symptoms, assessing information in real

time, and personalized recommendations. They are mostly repositories of Ayurvedic knowledge and not intelligent systems that can provide meaningful guidance [17].

2.3 Mobile Health Applications

With the advent of mobile health (mHealth), some of the wellness apps have added elements of Ayurveda. These applications, generally include Dosha quizzes, yoga routines, meditation tracks and a few basic lifestyle suggestions. While they do provide interactive interfaces, they are based primarily on predetermined questionnaires and less on dynamic interpretation of symptoms or Machine learning type of prediction. Their diagnostic models are often quite simplistic and rely on simple scoring mechanisms instead of robust computational logic processes. Moreover, lack of structured datasets, validated symptom-Dosha maps or AI-driven reasoning limits their accuracy in diagnosis. Users can get non-constant and generalized assessments because of the lack of true personalization. These apps tend to be more generic wellness applications as opposed to being effective scientific Ayurvedic diagnostic systems.

2.4 Digital Positive Ayurveda Platforms

In recent years, some digital platforms have come up aiming to replicate the consultation of Ayurvedic practitioners through platform interface. These systems usually consist of human practitioners providing video consultations, chat- and question formats. Although they do make it more accessible to people seeking Ayurvedic applications, the diagnosis process is still very dependent upon human interpretation and availability of practitioners. The lack of automation acts as a limiting factor related to scalability and adding costs to the operation. Furthermore, these platforms do not regularly apply machine learning models to aid in decision-making or analysing the symptoms. Consequently, they have not worked on the development of computational ayurvedic models and structured datasets, which aid **the progress of research in the field of** AI-driven Ayurveda [18].

2.5 Herbal product E-Commerce Platforms

A good number of digital places are dedicated to the sale of Ayurvedic herbal products, supplements, and wellness kits. These platforms often come with recommendation engines that will recommend products based on basic user preferences or trending categories. However, such recommendations are rarely this grounded in Ayurvedic principles of diagnosis or a personalized assessment. Instead, they rely on conventional e-commerce algorithms such as collaborative filtering or the popularity ranking.

There is little in the way of integration on authentic Ayurvedic logic (e.g., Dosha analysis, finding remedies based on symptoms), and These systems too add little to the accuracy of bringing a diagnosis to light, they often perpetuate commercial products, instead of solutions aligned to evidence-believing wellness.

2.6 Review of Research Papers

Academic work on connecting Ayurveda and digital technologies has been steadily increasing, indicating a lot of interest towards computational modelling of the traditional medical knowledge. Several works have been done to classify Prakriti using machine learning, analyse Ayurvedic texts using natural language processing and analyse diagnostic potential of computational tools. Research in Ayur genomics has also given benefits to science to learn about the relationship between Dosha types and genetic markers to show biological relevance of Ayurvedic classifications too. In spite of all these advancements, there is currently no unified framework on AI-driven Ayurvedic Diagnosis. Studies concentrating on the Prakrit classification tend to use questionnaires based on Ayurvedic theory, and cluster algorithms, support vector machines or decision trees to determine dominant Dosha type. While these studies are interesting, they are based on self-reported inputs instead of analysing the symptoms. Other lines of research look into the field of sentiment analysis, mining text and semantic interpretation of ayurvedic scriptures. Although such efforts have been done to show digitization of Ayurvedic literature, creating a standardized set of data and automated diagnoses has not materialized. Recent publications exploring Retrieval-Augmented Generation based methods have demonstrated encouraging results in fields that require knowledge extraction from unstructured texts, but these methods have not been widely used for the field of Ayurveda [19].

Therefore, the current scholarly work demonstrates a considerable gap between the theoretical and practical work. Most of the published studies either cover a limited aspect of Ayurvedic diagnostics or work in conceptual modelling rather than full-scale implementation. AyurSage tries to bridge this shortcoming by integrating methods like symptom extraction, Dosha mapping and machine learning with the latest user interface technologies using a unified framework for the diagnosis.

2.7 Presently Followed Methods of Dosha Classification

Dosha classification has undergone a number of academic and experimental models, but current approaches are limited. Traditional Prakriti classification tends to make use of questionnaire-based methods wherein individuals are asked to rate their physical, mental and behavioural attributes. These are then evaluated either manually or algorithmically to identify dominant Doshas. While

helpful, these approaches do not take into consideration symptoms, which are crucial towards understanding Vikriti or current imbalances. A system without the analysis of symptoms cannot make meaningful provisions for the diagnosis and personal therapeutic recommendations.

Some studies have discussed the machine learning models, where the questionnaires are used to train the machine learning systems with the help of which we can predict the Prakriti types. These models comprise algorithms such as logistic regression, support vector machines, random forests and k-means clustering. However, these systems seldom work with data from real-world patient symptoms because of the lack of a structured Ayurvedic clinical data. As such, the ability to use them practically in the field of diagnostics is limited. Machine learning models based on small questionnaires and often subjective ones cannot develop the necessary precision to perform reliable ayurvedic assessments.

Attempts to extract the symptoms from Ayurvedic texts have been undertaken using natural language processing techniques such as named entity recognition, keyword extraction and rule-based parser. These techniques suffer from poor accuracy when applied for complex and metaphorical ayurvedic terminology. More advanced forms of NLP like transformer-based models are providing relatively better understanding but require structured and annotated datasets which are still scarce [20].

2.8 Identified Research Gaps

A critical analysis of the existing systems and academic literature reveals multiple research gaps which justified the development of AyurSage. The biggest gap is related to the lack of structured datasets on Ayurveda. Without any curated symptom-Dosha mappings, machine learning models cannot be properly trained, which restricts the possibilities of getting accurate computational diagnosis. Traditional Ayurvedic knowledge is still tied in a form of unstructured text data which requires the use of advanced extraction techniques.

Another big gap exists in the absence of personalized Ayurvedic diagnostic tools. Existing systems either provide generalized information or are based on practitioner interpretation of results, neither of which are successfully scaled. There is definite need for computational system that will automatically extract, interpret and apply the Ayurvedic knowledge on real-time. Current digital Ayurveda platforms are not individualized with machine learning models which are capable for analysing symptoms and predicting the Dosha imbalance. Systems that do attempt to classify it often rely on self-reported questionnaires rather than on symptom-based evaluations, so such systems are not suitable for diagnosing Vikriti.

Furthermore, no platform currently combines data extraction, machine learning diagnosis, and modern UI technologies in a combination. A great deal of research studies stops at the conceptual level with no implementation. There is little use of vector database, semantic retrieval or models in Ayurveda. This lack of technological synergy means Ayurveda does not reap its full benefit of technological advancements such as digital health and artificial intelligence. AyurSage Mohamed Recognizing the empathy gap in intelligent systems, AyurSage puts forward the idea of a holistic pipeline from knowledge extraction, model training, frontend interaction on a foundation of Ayurvedic authenticity [21].

2.9 Summary of the Literature findings

The literature survey shows that although Ayurveda has gained a worldwide recognition for its holistic characteristic, its digital representation of Ayurveda remains underdeveloped. Existing systems do not offer diagnostic support, but rather offer informational or commercial content. Research studies delving into Ayurveda with the guidance of AI are encouraging but incomplete and lack complex datasets, implementable and scalable models. The lack of organized symptom and Dosha associations as well as reliance on questionnaire-based methods makes construction of effective computational diagnostic tools difficult. AyurSage stands out from the rest in terms of tackling these limitations using extraction, building the structured dataset, machine learning based classification and modern UI integration. The results derived from the literature bring out the need for a system such as AyurSage and confirms its role in this new area of work of AI-enabled traditional medicine.

Table 1: Overall Summary Table

Category	Description / Findings
Existing Systems Overview	Most existing Ayurvedic digital tools are limited to static information, basic quizzes, or e-commerce platforms. They lack personalization, AI-driven diagnosis, or symptom-based analysis.
Static Ayurvedic Information Websites	Provide generic descriptions of Doshas, diets, and remedies. No data processing, AI prediction, or personalized guidance. Mostly text-based informational platforms.
Mobile Health Applications	Offer wellness tips, yoga routines, general Dosha quizzes. They do not use machine learning, do not extract symptoms, and provide only pre-defined suggestions.
Digital Ayurveda Consultation Platforms	Involve human practitioners through teleconsultation. Not AI-based, no automated diagnosis, and no computational interpretation of symptoms.
E-Commerce Herbal Platforms	Focus primarily on product sales. Recommendations rely on product-based filters, not Ayurvedic diagnostic logic. No Dosha prediction or symptom analysis.
Research on Prakriti/Dosha Classification	Limited to questionnaire-based approaches using ML algorithms. Lacks real symptom datasets. Does not support real-world diagnostic prediction.
NLP Research in Ayurveda	Attempts made to extract terms from Ayurvedic texts using NLP. However, classical texts are metaphorical; extraction accuracy is low without advanced AI (e.g., RAG).
Identified Research Gaps	No structured Ayurvedic datasets; no AI-based Dosha prediction systems; lack of symptom-to-Dosha mapping models; no integration of retrieval + ML; minimal academic work on automating Ayurvedic diagnostics.
Relevance to AyurSage	Confirms the necessity of RAG-based dataset creation, ML-driven Dosha prediction, and a unified digital Ayurvedic consultation system. AyurSage fills all identified gaps.

Category	Description / Findings

Table 2: Paper wise Findings Table

S. No.	Author(s) & Year	Title / Study Focus	Objective / Purpose	Methods & Dataset Used	Key Findings + Relevance + Limitations
1	Jadhav et al., 2024	<i>Integration of Machine Learning in Ayurveda</i>	Explore ML for diagnosis, personalization & disease prediction.	Review of ML algorithms (SVM, KNN, RF, NB, NN, LSTM); literature & Ayurvedic texts.	ML improves diagnosis & personalization; no clinical validation; foundational for ML–Ayurveda integration.
2	Roy et al., 2025	<i>Explainable ML for Prakriti</i>	Predict Prakriti using psychometric traits with explainability.	RF, XGBoost, SVM, LR, MLP; SHAP explainability; n=373.	RF AUROC≈0.762; EQ→Vata, IQ→Kapha correlations; self-report & imbalance issues; supports behavioral+ML features.
3	Sudha et al., 2025	<i>Ayurveda Data Standardization for ML</i>	Standardize heterogeneous Ayurveda text for ML improvement.	NLP preprocessing, NER, CNN, NB, BERT; Kaggle dataset (~300).	BERT 100% accuracy; dataset small; Sanskrit complexity; necessary for structured datasets in Ayurveda ML.
4	Sharma, P., 2025	<i>Uses & Relevance of AI in Ayurveda</i>	Study how AI modernizes Ayurveda (diagnosis, personalization, research).	Narrative review of Ayurvedic texts + scientific databases.	Highlights AI's role in Prakriti ID, imaging, consultations; limited standardized datasets; supports AI-based Ayurveda systems.

S. No.	Author(s) & Year	Title / Study Focus	Objective / Purpose	Methods & Dataset Used	Key Findings + Relevance + Limitations
5	Ramesh, V., 2025	ML in Ayurinformatics	Explore ML for personalized care & herb-drug modelling.	Review of ML tools (TensorFlow, Spark, XGBoost, NLTK, etc.).	ML aids prediction & decision support; lacks real-world validation; relevant for AI-based Ayurveda systems.
6	Gupta & Mehta, 2025	AI in Ayurveda (Systematic Review)	Assess AI applications in diagnostics & prediction.	PRISMA review of 60+ studies (ML, DL, NLP datasets).	AI improves Prakriti/disease prediction; dataset gaps; useful for designing ML-Ayurveda models.
7	Sudha et al., 2025	Ayurveda Data Standardization	Standardize text for ML readiness.	NLP, NER, CNN, NB, BERT; ~300-entry dataset.	BERT 100% accuracy; small dataset; crucial for structured AI-Ayurveda data.
8	Gupta et al., 2025	AI Applications in Ayurveda	Review AI use in diagnosis & drug research.	Literature-based review (NLP, ML, imaging, genomics).	AI supports imaging & personalization; limited datasets; strengthens conceptual background.
9	Majhi et al., 2023	ML for Parkinson's via Dosha Features	Predict Parkinson's disease using Ayurvedic dosha correlations.	Models: LR, KNN, SVM, KSVM, NB, DT, RF, XGBoost; dataset n ≈ 80,916 (Fox Insight).	KSVM ≈93% accuracy; Vata traits linked to PD; lacks DL models; useful for combining clinical + dosha features.

S. No.	Author(s) & Year	Title / Study Focus	Objective / Purpose	Methods & Dataset Used	Key Findings + Relevance + Limitations
10	Katkar et al., 2025	Ayurveda Hub: Automated Prakriti Detection	Build AI-based prakriti detection + text reference system.	MERN + ML (SVM, DT, NN); NLP (TF-IDF, cosine similarity); user data.	Accurate prakriti detection; slow on large datasets; helps develop AI-Ayurveda diagnostic systems.
11	Akarasereenont & Jongjamdee, 2025	AI in Traditional Medicine	Review AI use across Ayurveda, TCM & others for diagnosis & personalization.	Narrative review (2017–2025); ML, DL, NLP examples.	AI improves pulse/face/tongue diagnostics; dataset scarcity; guides future AI-Ayurveda frameworks.
12	Ranade, 2024	AI in Ayurveda: Concepts & Prospects	Explore AI roles in diagnosis, drug discovery, and decision support.	Review of 34 AI-Ayurveda studies (from 110 screened).	AI aids prediction & research; limited dataset strength & validation; foundational for AI-Ayurveda tools.

CHAPTER – 3

DESIGN

3.1 System Architecture

AyurSage architecture is a developed system based on the integration of machine learning and modern web technologies provided in the form of a full-stack program designed to provide personal Ayurvedic health analysis. The system is modular and scalable in nature to achieve ease of data flow, efficient processing and ease of interaction with the users. It is designed in **three main layers** namely: data **layer**, processing **layer** and presentation **layer**. Every **layer** carries out a particular task and the whole structure is working together in order to achieve the overall functionality of the system.

The data layer is the base of the system and it is a structured Ayurvedic dataset, which is in CSV format. This database holds the data that relates the user symptoms, lifestyle parameters, types of doshas, diseases and the respective treatment like therapy, medicines, diet plans and exercises. There is a scarcity of standardized Ayurvedic datasets hence this paper utilized a curated dataset to both train and infer. Moreover, the trained machine learning models and label encoders are saved in the form of serialized (.pkl), making them easy to load and make predictions at any point in time. This layer will make sure that any knowledge that needs to be used in prediction and recommendation are easily available in a structured and accessible format.

The processing layer is in charge of receiving user inputs and performing machine learning predictions and getting results. This layer is composed of two key parts, the machine learning server and the backend server. The backend is created on a basis of Node.js and Express that handles user authentication, API routing, and interaction between the frontend and the machine learning server. Machine learning is applied with the help of Flask in Python on which trained classification models are applied. Upon loading input data which includes age, gender, symptoms, diet, sleep patterns and stress levels, the backend takes the input information and passes it to the Flask server through REST API calls. The machine learning models would then take the input to predict the predominating dosha imbalance and the disease. According to the prediction, the dataset retrieves the relevant treatment recommendations. This layer provides effective calculation and optimal integration of various technologies.

React.js is used to put the presentation layer (user interface of the system). It offers responsive and interactive features like the login/signup pages, assessment forms, result visualization pages, tracking history dashboard and consultation interfaces. The frontend accepts the inputs of the users and interacts with the backend via API calls. The findings obtained in the backend such as predicted dosha, disease, and treatment plan are presented in ordered and attractively appealing format. This layer is aimed at improving the user experience through a well-defined navigation, user-friendly design, and meaningful insights.

In general, this system architecture of the AyurSage is able to effectively combine machine learning with a full-stack web application to imitate Ayurvedic diagnosis. The system is reliable, is scalable, and practically usable by removing dependency on complex retrieval mechanisms, and concentrating on structured data and predictive modeling.

3.2 Workflow Diagram

The AyurSage workflow is a replicated simplified Ayurvedic diagnostics procedure by means of contemporary computational methods. The interface starts with the user by the front-end interface where the user enters the personal information and health parameters like age, gender, symptoms, diet type, sleep habits, stress, and seasonal. These are the main data to be analyzed.

After the user provides the information, the data will be relayed to the backend server that is developed with the help of **Node.js** and **Express.js**. The backend authenticates and formats the input **data** in order to make them consistent and then sends them to the machine learning server. This preprocessing step is used to make sure that all the required features are properly formatted in order to be inferred into the model.

The resulting processed data are sent to the Flask-based machine learning server by an API request. The machine learning models process the input and give the prevailing dosha imbalance and a likely disease with the presented symptoms and lifestyle factors. The system also determines probabilities of predictions to give levels of confidence. Upon prediction, the system recalls an adequate treatment plan within the dataset that comprises of therapy, medicines, nutritional guidelines and exercises.

The end result is relayed to the backend which in its turn transmits it to the frontend. The results are presented in a proper format in the frontend, such as dosha prediction, disease identification,

confidence levels, and customized Ayurvedic recommendations. The system also records the assessment data to the database making the users to be able to access their previous records through the dashboard.

This workflow provides a streamlined and effective data input and output as well as accuracy and usefulness. It removes the complexity that is not needed and concentrates on the straightforward Machine learning-based Ayurvedic analysis.

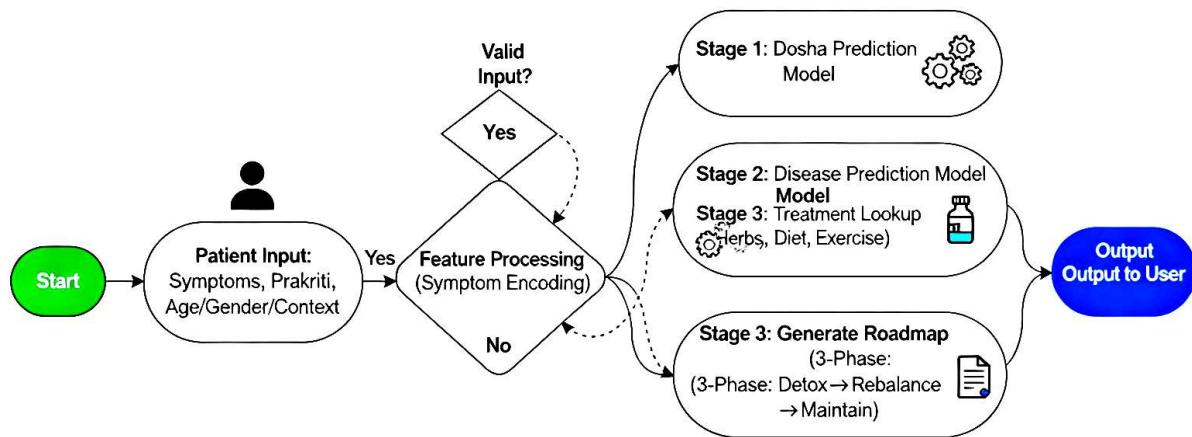


Figure 2: Workflow Diagram

3.3 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) is used to give a clear representation of the flow data in the AyurSage system and the interaction among the various components with one another. On the topmost layer, the system exhibits a nonexpedient interaction between the user and the application. The user enters details of the symptoms and lifestyle related to the information and the system processes it to give Dosha forecasting and customized Ayurvedic advice. Level 0 is a model that describes the system as a single process in which its user inputs are converted into meaningful outputs. The user is an outside entity, who interacts in the system by providing information in health-related information. The system then pushes the anticipated Dosha, potential disease and treatment recommendations back. Level 1 describes the working of the system internally in detail. After the user makes an input using the frontend interface, the data will be redirected to the backend server. Basic validation and preprocessing are done on the backend to verify that the input is made in the appropriate form. These involve the formatting of categorical values and the data prenatalization to fit machine learning models. The post-processing data is passed to the machine learning server that is

developed with Flask. The ML models work with input data and classify it to forecast the prevailing Dosha imbalance and the disease. Probability scores are also determined to show the accuracy of predictions. After the prediction is made, similar treatment recommendations are accessed in the structured data by the system. Such are the therapy, drugs, nutrition and physical activity recommendations. The outcome is later relayed back to the backend that communicates it to the frontend interface to be shown. Also, user information and prediction output are recorded in a database allowing the user to view past assessment in the dashboard. This hierarchical flow will guarantee the seamless flow of information between the frontend, backend and machine learning elements leading to an effective and stable system.

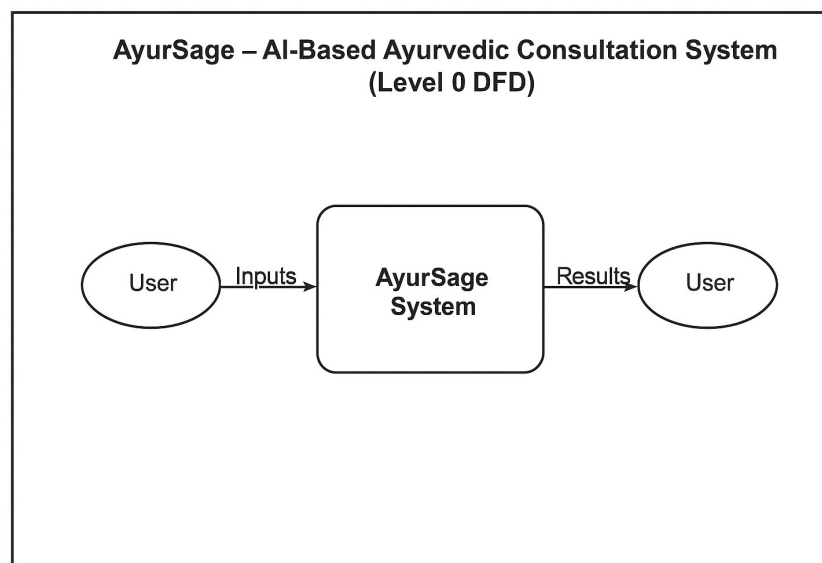


Figure 3: 0 Level dfd

3.4 SDLC Model Used: Agile Model

The AyurSage development is based on the Agile Software Development Life Cycle (SDLC) model, which is very suitable when dealing with a project of a continuous improvement and development cycle. The reason why agile methodology was selected is based on the dynamic nature of the system particularly in regards to refining the dataset, optimization of the model and increasing user interface. Agile is a development process that differs with the traditional Waterfall model as it has the ability of development in incremental cycles referred to as sprints. Every sprint is dedicated to a certain part of the system, i.e. frontend development, backend integration, or machine learning model refinement. This strategy facilitates constant testing, feedback and improving during the course of development. During the requirement analysis stage, it was determined that there was a need to combine Ayurvedic knowledge with machine learning methods. The system was created to make simulation of

Ayurvedic diagnosis based on structured data and predictive models. At the design stage, a modular architecture was established breaking down the system into frontend, backend and machine learning components. The implementation stage entailed simultaneous code-writing of modules. The machine learning models were optimized with a series of iterations and the frontend and the backend were constantly optimized as the results of the test. Integration testing involved making sure that all the components were operating together effectively. Agile also provided flexibility in dealing with changes, including the input parameter changes, better prediction logic and better user experience. This cyclic methodology made sure that the end product in terms of the system is accurate and user friendly.

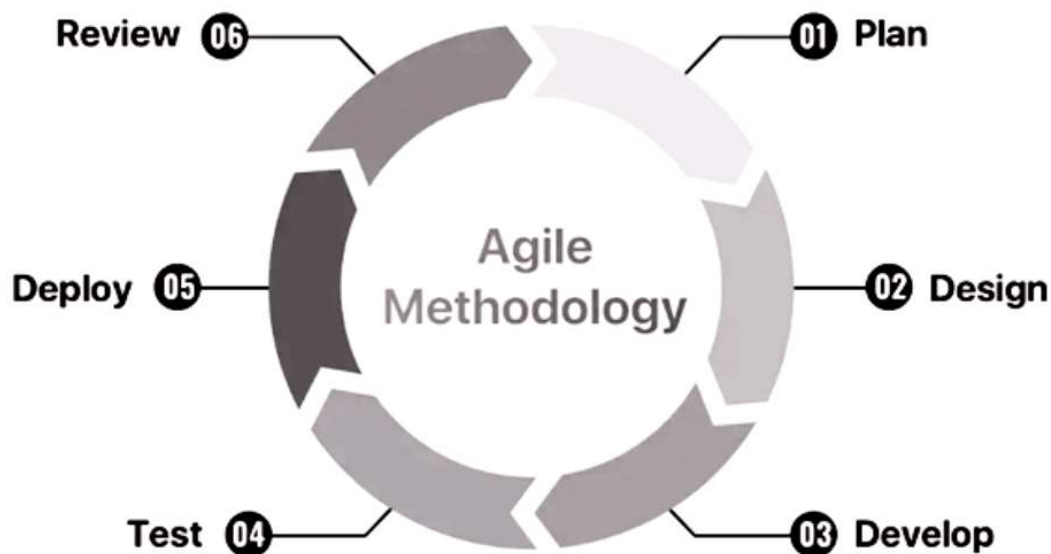


Figure 4: Agile Model

3.5 Use Case

The system functionalities are described in the use case framework of AyurSage. The user who consults Ayurvedic with the assistance of the application is the most important participant of the system. The user feeds on the system with inputs of personal data and information regarding his or her health including symptoms, dieting, sleep, stress, and other lifestyle parameters. The system processes the data on the basis of these inputs and gives predictions and recommendations. The system has a number of functions that it undertakes internally like data validation, preprocessing, machine learning prediction, and recommendation generation. The backend server and the machine learning server handle these operations and form internal components of the system. The most important applications of the system are: Authentication (sign up and log in) of users. Health lifestyle

data submission. Prediction of dosha and disease. Presentation of individualized treatment instructions. History of user assessment storage and retrieval. Seeing dashboard and consultation controls. All these applications together constitute all the functionality of the system and they are seen to show how AyurSage can give a digital replica of an Ayurvedic consultation process.

3.6 Gantt Chart

The creation of AyurSage at various stages was planned and followed with a Gantt chart. The project was further separated into several phases: the requirements analysis, system design, model development, the back-end implementation, front end development, integration, testing, and deployment. The first step was aimed at learning Ayurvedic and gathering system requirements. This was followed by the dataset preparation and developing of the machine learning model that included training, testing, and optimization. Versioning Backend and frontend components were developed in parallel. The server was built on the Node.js and Express.js, whereas the front end was coded on React.js. The integration was carried out once each of the modules was accomplished. During the testing and debugging, reliability and accuracy of the system were tested. The Gantt chart was useful in determining the task dependencies, progress, and the project completion. It also facilitated effective management of time and synchronization of various stages of development.

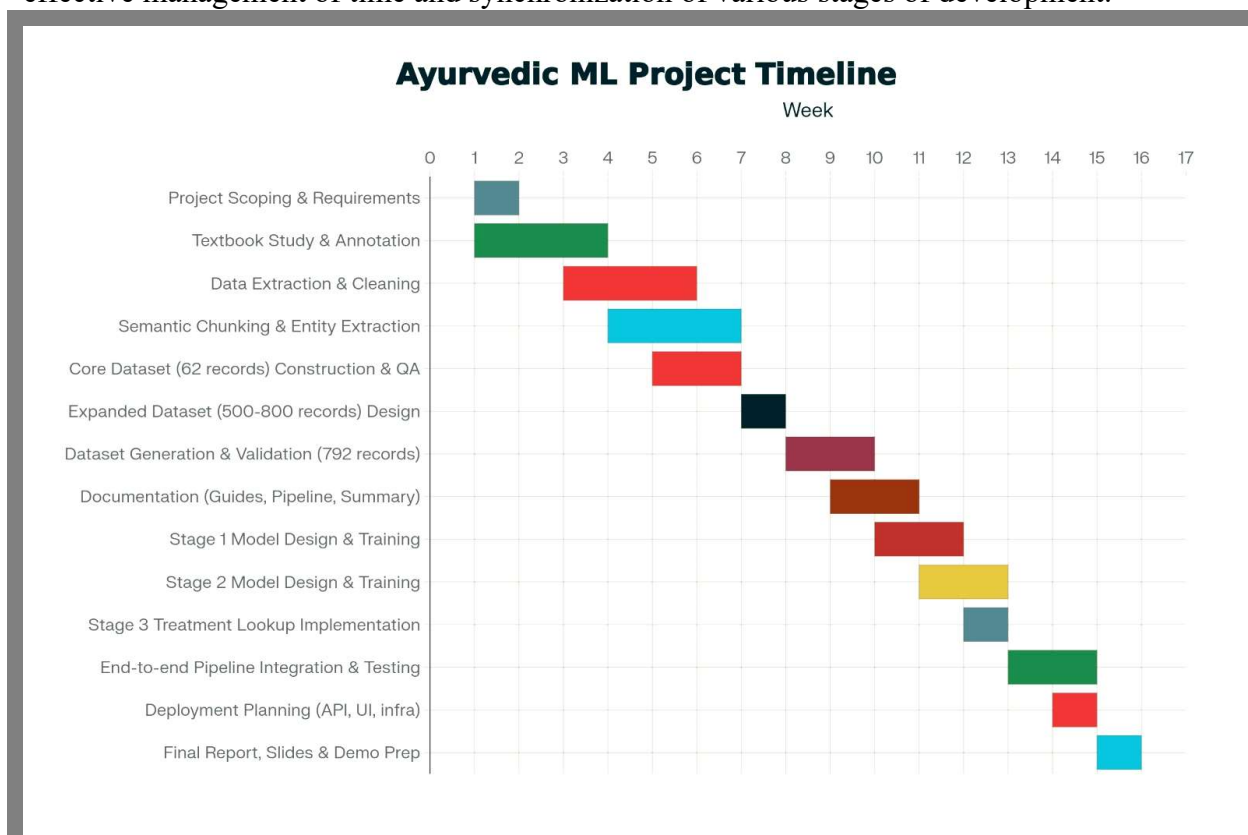


Figure 5: Gantt Chart

3.7 Components of the System Description

AyurSage is a combination of various interrelated elements that collaborate to mimic an Ayurvedic diagnostic system with the help of machine learning and web technologies. The machine learning module will be the first element, and it will predict the Dosha imbalance and potential diseases. This module involves the use of trained classification models that can process the user input, which includes the symptoms and lifestyle parameters. The models produce predictions and confidence scores that are helpful to know the quality of results. The second one is the dataset module that includes organized Ayurvedic data that is used in training and inference. This data contains correlative relationships among symptoms, doshas, diseases and treatment prescriptions in form of therapy, medicines, diet and exercise. It is the repository of the system knowledge.

AyurSage database layer is based on MongoDB Atlas, a cloud-based NoSQL document database, which offers the flexibility of the schema design, horizontal scaling, and high availability. The choice of MongoDB compared to the relational databases is due to the nature of the Ayurvedic health records which are inherently variable and multi-dimensional in nature, where different users enter varying combinations of symptoms, lifestyle factors, and personal information, and a set relational schema cannot be used in this case.

The back-end component is made with the help of Node.js and Express. It takes care of API routing, user authentication as well as communication between the frontend and the machine learning server. It secures and efficient information exchange in the system. The flask server is built with Python to execute the machine learning server. It does predictions and provides results in a format. The frontend system is developed on React.js and offers a user interface. It enables one to key in their information, see forecasts, and have personalized suggestions. It has also incorporated other features like dashboard and consultation interface to be able to provide better user experience.

Last, recommendation part produces individualized Ayurvedic recommendations depending on the prediction outcomes. It also offers treatment recommendations, nutritional guidelines, herbal supplements, as well as lifestyle recommendations, to aid users in regaining their health balance. These components create a seamless system that combines traditional Ayurvedic with machine learning and full-stack development.

CHAPTER – 4

METHODOLOGY

4.1 Data Mining (Data extraction PDF Parsing)

The methodology adopted for AyurSage starts by performing a huge data-extraction process, which is required because of the lack of any standardized and publicly-available data set in the field of Ayurvedic diagnostics. Since Ayurveda exists mainly in textual and unstructured form, namely, in the form of classical scriptures, research articles, and descriptive academic material, the first step of the methodological approach requires to systematically extract relevant diagnostic knowledge from such sources. Traditional ayurvedic manuscripts have elaborate descriptions of symptoms, dosha imbalances, pathological states and therapeutic interventions. However, such descriptions are not easily accessible in forms that can be used for machine learning. Therefore, AyurSage begins the process with the pdf parsing, in which digitized literature in Ayurveda is transformed to raw text type. The thus extracted text is pre-cleaned to eliminate noise, formatting artifacts and irrelevant content, while retaining the semantically rich Ayurvedic descriptions.

Once This project is built upon the basis of creating a clinically valid dataset by using classical Ayurvedic texts. Three credible books, namely Charaka Samhita, Sushruta Samhita and Ashtanga Hridaya, were used to extract primary textual data manually. These books were scanned and extracted to determine any pertinent content that talked about the state of the diseases, the dosha imbalances that accompany them, the symptoms of the diseases and the treatment that are prescribed.

The isolated clinical patterns of these texts were isolated using PDF parsing tools and manual annotations. All the extracted records have been validated with due care to maintain authenticity and support of relationships between disease, dosha, symptoms, treatment and dietary recommendations. The outcome of this process was 792 patterns of textually-supported Ayurvedic clinical entities.

4.2 Preprocessing of Data & Feature Engineering

Preprocessing the dataset in order to make it ready for machine learning. The extracted mappings often need normalization in regards to terminology variations, overlapping expressions, overlapping descriptions and contextual interpretations. Ayurvedic texts often use metaphorical language on describing symptoms or dosha behaviors which requires the process of semantic normalization to

gain consistency. Therefore, symptoms are standardized using a standardized vocabulary which is in line with the modern terminology of Ayurveda. Dosha labels are translated in to categorical values with descriptions that are not clear being resolved by cross-referencing multiple sources.

The textual data that was extracted was first preprocessed to make the data consistent and useful in machine learning. Preprocessing was done in a few stages: Data Cleaning: Elimination of redundant patterns, term standardization, and dosha, symptom, and disease names normalization in order to have a consistent set of labels. Data Structuring: Transformation of text-based patterns into structured tabular data having specific feature fields. Feature Engineering: 16 features were determined based on the contextual and clinical features. These are demographic (age, gender), lifestyle (diet, sleep quality, level of stress, season factors, etc.) and clinical (pain, digestion, energy, temperature, skin condition, etc.) symptoms. Label Encoding: The dominant dosha (Vata, Pitta, or Kapha) and Ayurvedic disease classification (22 in total categories) were transformed into target variables.

Feature engineering is a very important step in improving the model. Ayurveda does not only look at the symptoms but rather several contextual factors like age, gender, sleep cycle, stress, eating habits, seasonality, etc. while determining the imbalance of Doshas. These features were implemented into the dataset in a structured effort to get a good set of predictors. Since Ayurveda considers the presence of certain tendencies (Vata predominance for elderly people or kapha imbalance for sedentary habits) which are contextual, these real-life features enhance the power of prediction of machine learning model.

4.3 Mapping of Symptoms to Doshas Logic

does not have explicit presentations of the relationships between symptoms and doshas in a tabular context; rather, such relationships are contained in reports of narratives, guidelines for therapies, and philosophical explanations.

Both records of the dataset relate certain symptoms to the prevalent doshic imbalances on the basis of textual accounts. The logic of mapping makes use of the Ayurvedic rules:

Vata Dosha: Dryness, irregular appetite, constipation, anxiety, and hysterical energy are some of the symptoms that are related to this Dosha.

Pitta Dosha: This is usually associated with heat, inflammation, anger, excessive hunger, skin rash.

Kapha Dosha: This is the tendency of being heavy, lethargic, congested, and the food is slow to digest. When creating the datasets, the matching of symptoms to their dosha was performed based on

a manually curated symptom-dosha dictionary that was generated based on the classical sources. This interpretation methodology makes sure that all the dosha predictions that the model will do at a later stage can be tracked down to definite, textually validated associations.

4.4 Creating and Validating Datasets

Dataset creation is an important step in the AyurSage methodology as the system is based on a structured and validated dataset to train the machine learning model. The dataset has several types of features, such as main symptoms, lifestyle factors, demographic information, and contextual factors. Once the data set is assembled, the data is then validated for quality and accuracy. Validation is done with consistency check to identify contradictory mappings, frequency analysis for finding anomalous patterns and domain verification with classical ayurvedic principles. Outliers or inconsistency is fixed or eliminated. Further, the data set is statistically analyzed to detect imbalances in classes, a fact that frequently happens when the types of Doshas predominate in the mapping over others. Techniques like oversampling, under sampling or synthetic data augmentation might be used to reduce bias. A clean and validated data set is the end result, which is ready to use for training a model.

The process of dataset creation had two major stages: Phase 1: Core Pattern Extraction: There were 792 canonical patterns of patient generated directly out of the canonical texts. Both patterns explain the combination of disease, the dominating dosha, primary and secondary symptoms, treatment and lifestyle factors, each of them being backed by clear textual references. Phase 2- Controlled Augmentation: Such base records were extended to about 3,000 patient-level data points by different contextual qualities like age, sex, prakriti (body constitution), climate, and season, and clinical areas remained unchanged. This method had realistic diversity, but did not incorporate any artificial or made-up clinical information. Validation: a sample set was reviewed by Ayurvedic domain experts to ensure semantic accuracy and make sure that augmented records made clinical sense.

4.5 Model Selection & Training

Model selection is an important step in the construction of a correct and reliable dosha prediction system. The aim is to test different machine learning algorithms and identify one that captures the complex relationship existing between the symptoms, contextual features and dosha imbalances. Several models were experimented, such as logistic regression, decision trees, random forests and support vector machines. Initial baseline models were used to provide a performance benchmark and more sophisticated ensemble approaches were used to improve the accuracy.

The model selection and training are carried out as follows: Two machine learning models based on ensemble were trained: Dosha Prediction Model: This is a Random Forest Classifier used to predict the leading dosha in terms of patterns of symptoms and contextual characteristics. The random forest classifier showed an excellent performance given its capabilities of coping with non-linear interactions between features, multi-dimensional data structures, and categorical variables. It helps in mitigating overfitting by ensemble averaging and gives the feature importance that can be interpreted by visualizing it. The interpretability and strength of the model render it appropriate in areas of application that involve trust and explanation in the healthcare field. Disease Prediction Model: The input features such as primary symptoms, dosha type and contextual indicators were used to predict one of 22 Ayurvedic disease classes using an XGBoost Classifier. The data was separated into training (80) and testing (20) data sets. The grid search was conducted to hyperparameter optimizing on accuracy, precision, and recall to get a balanced performance on all categories. Evaluation metrics like accuracy, precision, recall and F1-score were utilized to evaluate performance. Cross-validation techniques were used to further validate the ability of the model to generalize on different subsets of data. After satisfactory performance was obtained, the trained model was serialized with joblib for deployment in backend. This trained model is transformed into the predictive model, with the dosha prediction capability in AyurSage giving it instant prediction opportunity, once it gets a new input data set.

4.6 Evaluation Metrics Used

The performance of the Dosha classification model needs to be evaluated by powerful statistical measures that encompass both accuracy and reliability. Accuracy gives a general measure of performance being the ratio of correct predictions. However, as Dosha categories may not be uniformly distributed, the accuracy is also not sufficient. Precision is the measure of how accurately a model is able to accurately predict a particular Dosha when it says that particular Dosha. Recall assesses the ability of a model to recall all the relevant cases for a given Dosha, such that it is key to diagnosis of an imbalance. The F1-score is a way to get a balance between precision and recall in order to evaluate equally between classes.

Confusion matrices were applied for visualizing the frequency of misclassification of symptoms between Doshas by the model. Since Vata, Pitta and Kapha symptoms also sometimes overlap, the confusion matrix gives some insights into where the model may need further tweaking. Cross validation helps to check whether the model performance is the same over different subsets of the

data. These evaluation metrics taken together verify that the model is reliable, generalizable and can be deployed in the real-world.

4.7 Recommendation Engine

The system has a rule-based recommendation engine, which comes up with Ayurvedic lifestyle and treatment recommendations depending on the diagnosed disease and dosha imbalance. Appropriate therapies, nutritional changes, herbal preparations, and daily lifestyle instructions are given in relation to each prediction. These suggestions have close connections with the textual sources, on which the clinical patterns were based on, which remain interpretable and traceable. The machine learning models and recommendation system are integrated with the help of a Flask backend, and the data is processed and stored in the database of MongoDB. Modular design will ensure that it can be expanded in future with new disease conditions or Ayurvedic domains.

Beyond the prediction of Dosha imbalances, the solution of AyurSage has a recommendation engine, which aims to deliver the users with practical guidance that is Ayurvedic. The recommendation engine is based on a curated knowledge base of Ayurvedic remedies that are related to each of the three Doshas. When the ML model rigs the dosha imbalance to recommend according to the prediction, the recommendation engine goes through its knowledge base to come up with relevant dietary recommendations, herbal suggestions, lifestyle corrections and seasonal precautions for that specific dosha imbalance.

For example, if the user has a Pitta imbalance the system advises using cooling herbs such as aloe vera or coriander and dietary advice such as drinking more water and avoiding spicy foods. If the imbalance is Kapha related, the system recommends warming herbs, light meals and increased physical activity. The recommendation engine is responsible for making sure that AyurSage works not only as a diagnostic system, but serves as a support wellness-companion, assisting users on their way to regain their balance. The workflow is also an imitation of the way an Ayurvedic practitioner combines diagnostic understanding and unique therapeutic counsel.

4.8 Summary of Methodology

The developed methodology is a systematic combination of the Ayurvedic textual information on one hand and modern computer tools on the other hand in order to derive a clinically dependable decision support system. Data mining was done in a rigorous manner with the identification, validation and translation of data into machine readable form, which allowed the creation of ensemble learning models that are able to predict dosha and disease states with high fidelity. The

system is applicable in the real-world Ayurvedic consultation and education due to its combination of structured dataset design, interpretable predictions and integrated recommendations.

The methodology of AyurSage combines the power of artificial intelligence, classical Ayurvedic and modern software engineering. Through the method of PDF parsing the unstructured Ayurvedic knowledge is converted into a recognition structure of Sonification knowledge for machine learning. Data preprocessing and feature engineering help in representing the symptoms and the contextual variables in a proper manner. With the help of this enriched dataset, the machine learning models have been trained to make accurate Dosha predictions. The recommendation engine gives a real-time ayurveda tips and frontend (React) and backend (Flask) help to ensure frictionless integration for user interaction in the system. This methodological framework makes AyurSage a technologically advanced and Ayurveles system.

CHAPTER – 5

IMPLEMENTATION

5.1 Implementation of the System Overview

The System Overview should be implemented in stages, starting with the initial phases to be conducted. The implementation of AyurSage is a combination of classical Ayurvedic sciences, artificial intelligence and more recent web application technologies within a single computational system. Although the conceptual basis is based on the traditional Ayurvedic principles, the application will translate the ancient knowledge into a functional digital platform that can offer Ayurvedic consultation and decision support in real-time.

The implementation stage is where all main elements are integrated the data extraction pipeline, machine learning models, back-end processing logic, and the frontend platform are implemented. The main architectural principles of modularity, scalability, and maintainability have been stressed so that each of the individual components can be developed independently of the entire system.

It starts with creation of organized data based on authoritative Ayurvedic writings (Charaka Samhita, Sushruta Samhita and Ashtanga Hridaya). This is a dataset consisting of disease-dosha-symptom-treatment relationships, which are used as the training basis of two classifiers, the Dosha Prediction Model and the Disease Prediction Model. The two models rely on the ensemble machine learning algorithm (Random Forest and XGBoost) to make interpretable and accurate predictions.

The Flask backend does all the processing and model-serving operations and exposes RESTful API endpoints between the trained models and the structured dataset. The React.js frontend offers a responsive and easy-to-use interface where the users can enter their symptoms, get predictions, and see lifestyle recommendations which are based on Ayurvedic logic. MongoDB is used to store data and model outputs, so that patient records, model results and recommendation mappings are flexibly and document-oriented stored. Each and every component of the system communicates with each other through clearly defined APIs and the separation of data, business logic, and presentation layers are maintained. With this implementation, AyurSage is able to combine the Ayurvedic diagnostic reasoning which has been tested over time and the modern software engineering practices to provide an interactive, trustworthy and extensible intelligent health assessment system. The AyurSage system is primarily centered on the dataset, which is carefully edited because there are no publicly available repositories of Ayurvedic data. These extractions generate 792 canonical clinical patterns that are the

fundamental records of the data set. To render the data representative of actual diversity in the real world, the contextual attributes, including age, gender, season, prakriti (body constitution), diet type, stress levels, and sleep quality are effectively varied to generate some 3,000 patient-level records. These additions are set to maintain the semantic and clinical consistency of the original patterns and bring demographic diversity.

5.2 Preparing and Preprocessing of Dataset

The dataset is the backbone of any machine learning system and in the case of AyurSage, it is especially important as there is no pre-existing dataset of ayurveda available publicly. The preprocessing is conducted in several steps to provide the data with the high quality of data and its use in machine learning applications: Data Cleaning: Identification and elimination of duplicated or conflicting records, standardization of textual words, and elimination of synonym-related variations in the name of symptoms and diseases.

Once the raw dataset is filled, the data preparation starts, this involves cleansing the dataset from inconsistent data, eliminating unclear mapping and normalizing the terminology. Any symptoms that also realize one equally with several spellings or synonyms show standardized to underlying conditions as much consistency as possible. Dosha labels are encoded into categorical values which are suitable for ML algorithms. Numerical features such as age are scaled to achieve uniformity and categorical features such as gender, type of diet, and quality of sleep are encoded using label or one-hot encoding methods. Missing values are addressed by either imputation or deletion of harsh data depending on the situation. The labels of dosha and disease are transformed into numerical values to fit machine learning. Scaling: Continuous data such as age and stress levels are scaled so that there is consistency in the data range.

Feature engineering is incorporated to give better predictive ability to the machine learning the model. Ayurvedic diagnostic reasoning is also affected by multifarious and complex interactions among symptoms and contextual variables; hence, composite features are formed in order to capture these multi-relationships. For example, "season x symptom" is being treated as a compound feature because some symptoms are captured more strongly in some seasons. The dataset is divided into training and testing subsets in order to help evaluate the model. The preprocessing pipeline is therefore an important way of ensuring that the dataset is accurate, thorough, and optimized for the machine learning tasks.

The preprocessing is conducted in several steps to provide the data with the high quality of data and its use in machine learning applications: Data Cleaning: Identification and elimination of duplicated

or conflicting records, standardization of textual words, and elimination of synonym-related variations in the name of symptoms and diseases.

5.3 Implementation of Symptom to Dosha Mapping

One of the most innovations in AyurSage is the implementation of symptom to dosha mapping. The Symptom-to-Dosha Mapping System is one of the main implementation aspects of AyurSage, providing the direct association between the symptoms, as it is entered by the user, and its corresponding doshic pattern according to the codified Ayurvedic principles. AyurSage uses rule-based mapping mechanism, based on canonical Ayurvedic sources instead of using text retrieval or external models.

Symptoms are associated with one or more types of Dosha: Vata, Pitta or Cough. This table was built in the extraction of data when verses and descriptions in core texts were analyzed. Once the user enters the symptoms by using the frontend interface, the backend uses this mapping logic to generate dosha probabilities. The mapping is a pre-classifier, which produces weighted Dosh distributions, and these are the inputs in the dosha prediction model. This is an amalgamation of manual rule-based assessment and machine-learning enhancement to guarantee not only the explanation but also the predictive power. Such ambiguities as overlapping symptoms are resolved by weighted averaging of the dosha possibilities based on statistical co-occurrence in data set. This makes identification of dosha to be consistent and context sensitive according to the Ayurvedic logic. The results of this mapping process give a clean and feature-based representation that is ready to be inputted into the model as well as interpreted.

The Symptom-to-Dosha Mapping implementation combining empirical textual reasoning and data optimization proves to be the basis of the diagnostic accuracy and human interpretability-critical requirements of Ayurveda-computational Ayurveda.

5.4 Implementation of Machine Learning Model

The machine learning model plays the role of the predictive engine of AyurSage. It is responsible for interpreting the symptoms provided by the user and takes into account the context to figure out which dosha imbalance is the most likely. The implementation starts by choosing the right algorithms depending on the characteristic of the data set. After experimentation with different models like SVM, logistic regression, and decision trees, it appeared that the most effective was the Random Forest classifier, because of its capacity to manage non-linear relationship, categorical variables and feature interaction.

Training the Random Forest model consists of taking the pre-processed data set and feeding it to the classifier and letting it build many decision trees. Each tree analyses other combinations of symptoms and characteristic features of the situation and learn patterns associated with Vata, Pitta and Kapha imbalance. During the training part, hyperparameters such as number of trees, tree depth and minimum samples to split could be tuned using cross validation methods to optimize the performance. The model is tested on the use of various metrics like accuracy, precision, recall, f1 score and confusion matrix analysis to identify the effectiveness of the model in all the three categories of Doshas.

Once the model gives good accuracy, the model is serialized using joblib library and integrated into the backend. During prediction, inputs provided by a user are sent to the same preprocessing pipeline of the training to have consistency. The processed data is then passed to the ML model which provides the predicted output of the Dosha in real-time. This prediction is the basis for the generation of Ayurvedic remedies. The responsiveness and reliability of the ML model is what makes it a key part of AyurSage's intelligence-based consultation engine.

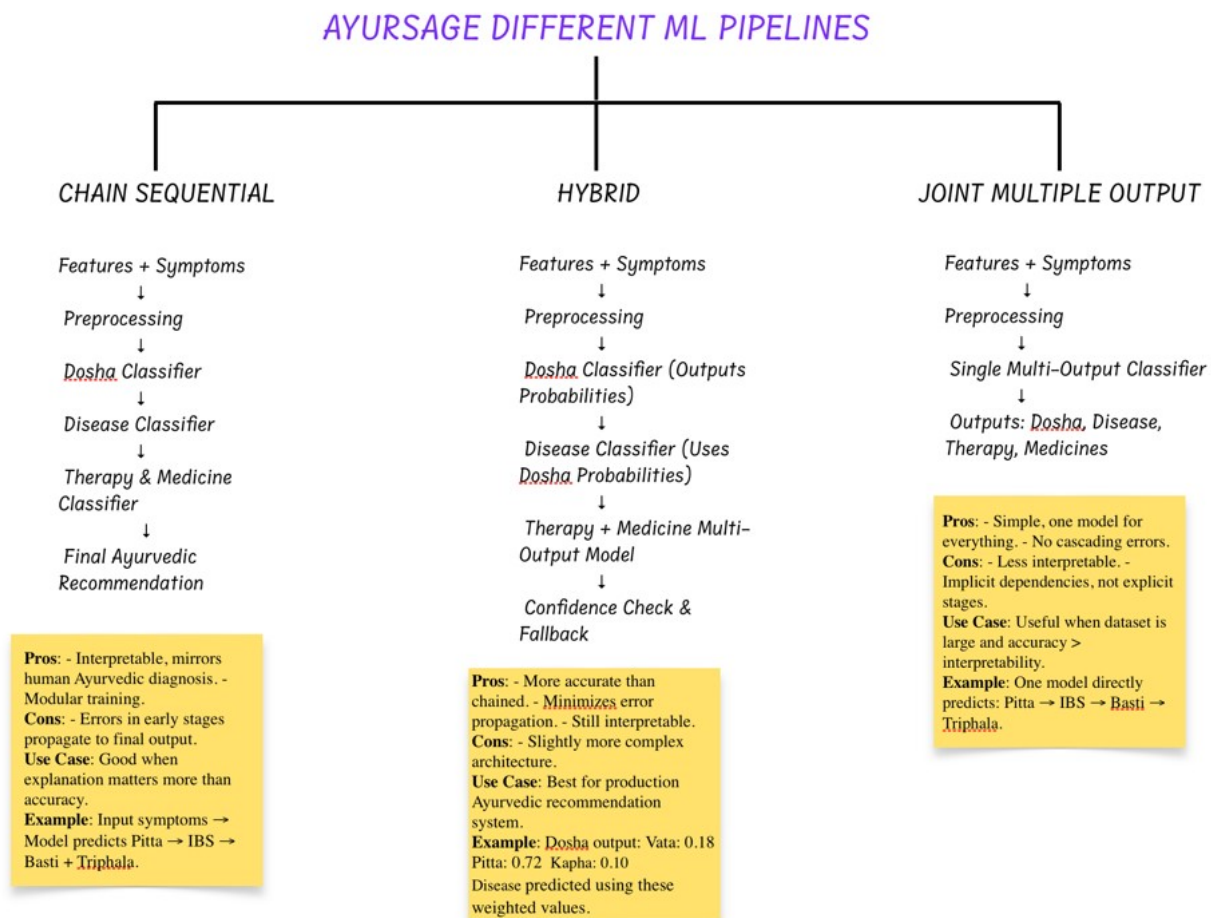


Figure 6: ML Pipelines



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5.5 Backend Implementation (Flask API)

The backend implementation is the core communication bridge between the frontend, the RAG engine and the machine learning model. Flask was used for backend development because it is light, flexible, and it is used for application programming interfaces. The backend exposes a few endpoints such as /predict endpoint to predict the Dosha and also others for user authentication, getting data and accessing a recommendation etc.,

When inputs are sent by the frontend to the backend then the servlet processing the data runs with the flask server and first of all transforms the user inputs coming from the client in a constant processing pipeline. The /predict handler loads the ML model already trained and the specially processed data and passes it on to make some predictions. The prediction output is then augmented with individualised recommendations of Ayurveda using extracted knowledge from the curated knowledge base. In case of cases that require specialized interpretation of symptoms, corresponding backend is responsible to trigger RAG pipeline to fetch relevant ayureckilled passages before giving the final output.

Backend implementation also includes the error handling, validation checks, and formatting the response to the frontend in order to interact with the frontend. The responses made by the API are in a format of delivered information (JSON) to facilitate the use with React. The modular nature of the backend enables different components like ML model, recommendation engine, RAG extractor, etc. to work individually while serving in conjunction with a unified architecture. There are security measures, such as CORS dealing and token-based authentication implemented where required.

5.6 Frontend Implementation (React.js)

The frontend of AyurSage is built by using React.js, a component-based JavaScript library known for its efficiency, scalability and easy development patterns. The frontend is charged with gathering inputs, initiating consultations as well as presenting results through visually appealing interfaces. The user experience, simplicity and responsiveness are prioritized in the design.

The workflow of the consultation starts with a multi-step form that involves patients in entry of symptoms and lifestyle information. Each form segment is represented as its own React component - it is possible to update and render it dynamically. React state hooks are function hooks that are used to handle user input and effect hooks that are used to handle asynchronous operations related to API calls. Buttons and navigation components to support smooth movement from one consultation step to the next.

After the user submits the consultation form frontend makes the API call to the backend. While waiting for the response, for a better user experience, some loading animation or spinner is presented. Once the backend transmits back predictions and recommendations, the frontend takes them and shows them in a visually structured manner. The result of the dosha is often shown with the color-coded cards of Vata, Pitta, and Kapha which come with Ayurvedic tips, dietary tips, and tips for remedies and sections. Additional pages like landing page, dashboard, about etc. are created using the React router so that seamless navigation is present in the Single page application.

5.7 UI/UX Design Principles Used

The user interface and user experience design principles applied in AyurSage emphasize on merging ayurvedic visual themes with modern web themes. The overall design features soft colour palettes which are inspired by nature such as earth tones for Kapha, warm colours for Pitta, and airy colours for Vata. Subtle gradients and shadow effects are employed as well as rounded corners to make the reading more calming and easier for the reader to understand.

User experience design has a focus on simplicity, clarity, and accessibility. The consultation form does not overwhelm the user with a lot of information and has been broken down into clear and manageable steps with instructions and logical design. Visual feedback mechanisms like hover effects, transitions and animated icons support interactivity and engagement of users. Animations are by choice subtle and that animation will not detract from the main functionality.

Typography is a key feature in improving the readability of the website where headers will be bolded with the use of modern fonts for the header section while the body text will be available with very legible styles. The UI caters for different familiarity level of Ayurveda to the users, while keeping the appearance of the system professional and yet approachable. These UI/UX principles play a very important role in ensuring that AyurSage is not only functional but also visually appealing to ensure user trust and comfort.

5.8 System Integration

System integration entails integrating the dataset, machine learning models, server of a backend, and the interface of a frontend into a single ecosystem. The initial stage of integration would be to make sure that the preprocessing pipeline that was applied in training the model is precisely recreated in the backend API, such that whatever feature engineering steps have been used during training data replicated during prediction time are applied to the live user inputs. The training and inference

pipelines need to be consistent to prevent the generation of malformed feature vectors and the poor performance of the models, and hence, this consistency is considered a critical integration requirement. The trained ML models are stored as serialized objects with the joblib library and loaded into the Flask server during startup and they are kept in memory and are available to serve prediction requests without loading overhead associated with making each API call.

The frontend would be connected to the Node.js backend through asynchronous HTTP requests, handled by Axios. When a user fills in the Dosha Prediction form and submits his or her inputs, the React frontend will build a structured JSON payload with all form fields and forward it to the backend through a POST request with the user JWT token on the Authorization header. The backend authenticates the token, retrieves the user identity and continues with the structured input to the Flask machine learning server through an internal Axios POST request. The Flask server takes the input in the preprocessing pipeline, runs the input in the Dosha Prediction Model and the Disease Prediction Model respectively and provides a JSON response with the predicted Dosha, the predicted disease, and the scores of the predicted disease and Dosha. This result is then saved on the backend in MongoDB in the collection of assessment history of the user and the entire prediction response is emitted by the frontend and rendered to the frontend. The general integration architecture adheres to a separation of responsibilities in three levels.

React frontend only deals with user interaction, state management, and presentation of the result. The Node.js backend will perform the role of authentication, data persistence, business logic and coordination of calls to machine learning server. The Flask server will only do model inference and send raw prediction results without any information of the user or the identity or the session context. This three-level distinction guarantees that the individual components within the system can be tested, debugged and scaled independently without interfering with the rest. The load on the machine learning server, such as the load, can be managed without the web server, and the database layer can be scaled with built-in horizontal scaling features of MongoDB Atlas without the need to make changes to the application code. With such a combined and stratified design, AyurSage is able to establish an effective, secure, and coherent workflow that unites AI-based Ayurvedic intelligence with a systematic and easy-to-use user experience.

5.9 Testing & Validation

Testing and validation are the keys to AyurSage ensuring it is reliable and accurate. Machine learning model validation is performed with the help of train-test splits, cross validation, confusion matrices and performance metrics. The backend is tested with tools like postman to verify the inputs

are handled correctly, the responses are structured correctly and the error management is handled correctly. Generates the backend logic is put through unit testing to validate each individual logic. Frontend testing encompasses testing the forms validations, user navigation, and API integration, and the responsiveness of the web pages for different devices. End-to-end testing is a series of tests to simulate a full consultation with a user to be sure that all components are working properly (from input to prediction to UI rendering). Usability Testing for Usability Testing is a testing that collects the views from users in order to make the workflow process and the interface better. Together, these testing methodologies make AyurSage run reliably and provide accurate Ayurvedic insights.

CHAPTER – 6

RESULTS

The AyurSage system is expected to bring a good, accurate and easy to use ayurvedic consultation experience through its combination of Machine learning and a modern react interface. The results are both a technical performance of the system and a usability of the system as a tool for an Ayurvedic wellness guidance.

6.1 Accurate Dosha Prediction

One of the main anticipated accomplishments of ayursage is being able to accurately predict an individual's dosha abnormality on the basis of the reported symptoms and contextual lifestyle factors. Traditionally Ayurvedic diagnosis has required the expertise of a trained practitioner, as based on the assessment of symptoms, behavior, physical attributes, environmental influence, conclusions are drawn about the state of the patient with respect to doshas. AyurSage seeks to replicate this diagnostic reasoning using a machine learning model using a structured dataset which is built from classical knowledge of Ayurveda. The correctness of the system is expected to be high because the dataset contains genuine symbol mappings of symptoms - Doshas based on Ayurvedic literature. By assessing subtle combinations of the symptoms, age, seasonality, diet, and the psychological factors, the nuanced model is expected to give reliable Dosha prediction. This result has great potential in making ayurvedic diagnosis accessible to people without access to trained practitioners or those who want immediate, preliminary diagnosis about their health imbalance.

6.2 Reliable Dataset

Another significant anticipated product of AyurSage is developing a dataset of symptoms, dosha imbalances, and other reliable structural science that is scientifically structured and of high quality. Currently, there is no public dataset available in the field of Ayurveda which captures the information due to the diagnosis in a machine-readable format. AyurSage Institutional Framework The resultant data set is expected to be coherent, standardized and reproducible and that in the future researchers can explore computational models for Ayurvedic diagnostics. The dataset will be of great significance from an academic perspective as it makes the centuries-old reasoning patterns of diagnosis codified at the data point, while the contextual meaning that is embedded in the texts is not lost. This dataset is not only the foundational training step on how to build the AyurSage model, this

dataset may serve another purpose for promoting digital Ayurveda - supporting further experimentation, AI research, and health informatics experiments.

6.3 Personalized Ayurvedic Recommendation

AyurSage is expected to provide customized ayurvedic recommendations to suit the predicted dosha imbalance. These recommendations range from herbal remedies, dietary remedies and lifestyle modifications based on the classical Ayurvedic therapeutic guidelines. Since the focus in Ayurveda is on individualized treatment, in this ideology, the system tries to imitate it and matches each Dosha with specific actions that result in the restoration of the balance. For instance, those presenting Pitta imbalance might get dietary advice to eat cooling foods and herbs with anti-inflammatory properties; whereas, those with Kapha imbalance might be advised to start adopting invigorating physical activities and warming food materials. These customized recommendations are real Ayurvedic therapeutic logic and helps the user to have a better understanding for the restorative health practices on the whole. Through such tailored guidance, ayurSage offers a meaningful connection between the diagnosis and the therapeutic action which helps in taking both prevention as well as action.

6.4 Interactive Web Application Interface

An intuitive and interactive user interface is one of the important expected results of AyurSage. The system is designed to provide a seamless consultation experience to the users with a clean, responsive and aesthetically pleasing React based frontend. The interface provides a structured multi-step submission form to give the user rate user-friendliness while submitting symptoms. The results of all the predictions are visually organized and are presented in form of Dosha interpretation, explanation, and suggested measures/methods to be adopted by the person. The aesthetic of the interface comes integrated with subtle Ayurvedic thematics through color palettes, gradients, typography, nature-inspired restless illustrations with traditional well-being concepts. Furthermore, we have dashboards in the system to track personal health trends, which will positively impact user engagement and long-term monitoring. The interactive interface also ensures that the technical complexity that is happening in the backend is invisible to the user, so that the user can focus only on a path towards wellness.

6.5 Increased User Awareness of the Ayurvedic Health

AyurSage is expected to add significantly to the public awareness and apprehension of the Ayurvedic health principles. By getting involved with the system the users gain insights about what the Dosha

imbalance signifies, the correlation between life style factors and health and the importance of preventive care. The system is designed to educate the users on how microscopic variances in diet, sleeping or emotional state, environmental, can affect their Dosha equilibrium. This awareness gives power to the users to lead a healthier routine as per the recommendations of Ayurveda towards leading in the path of preventative healthcare rather than the reactive path of treatment. The educational value of AyurSage is not limited to its use for diagnostics but helps in bringing about a more in-depth understanding of holistic wellness to the users who would otherwise not have been familiar with Ayurveda.

6.6 Practical Use Cases

AyurSage is anticipated to have diverse practical applications in the individual and institutional set ups. At individual level, users can use the system as a self-assessment tool for better understanding of their health patterns and taking preventive measures as per the logic of Ayurveda. At the wellness industry level, AyurSage can aid digital clinics, health coaches, yoga centers and wellness spas in the wellness industry with preliminary assessments to clients. In the academic factor, the fashioned data set established with AyurSage can be a tremendously useful facet for educationalists and tests for extra examining into computation methods dealing with Ayurveda. The mechanism of the RAG-based extraction may be adapted to other traditional system of medicine like Siddha, Unani or Traditional Chinese Medicine, thus expanding its effects. Moreover, AyurSage can support healthcare practitioners as they provide supplementary information that aids in the personalized treatment planning approach. These various use cases for practical applications illustrate the broad use and value to society for the system.

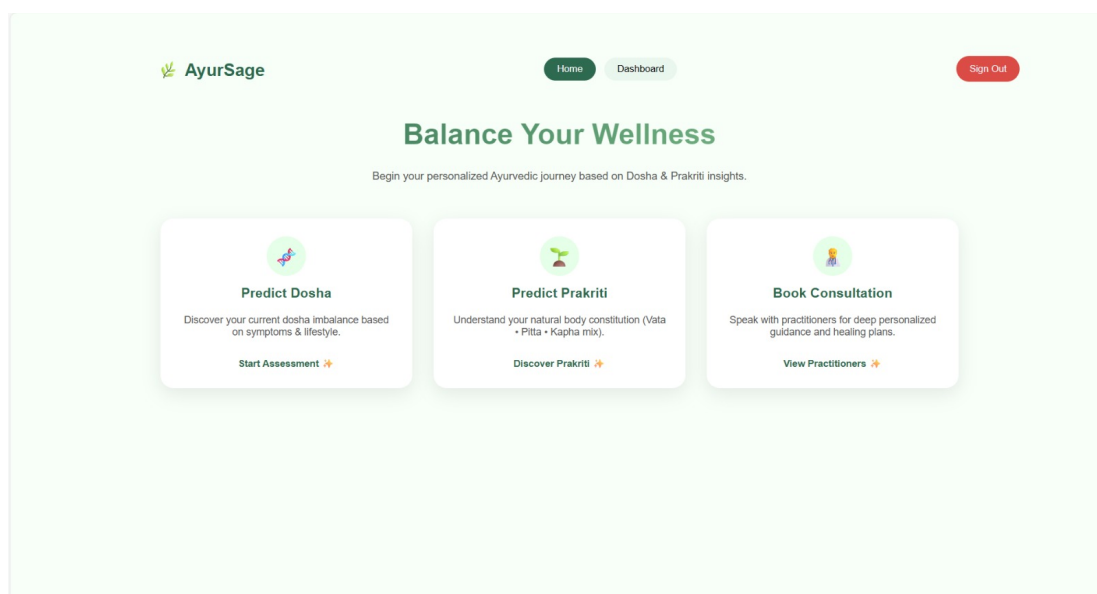
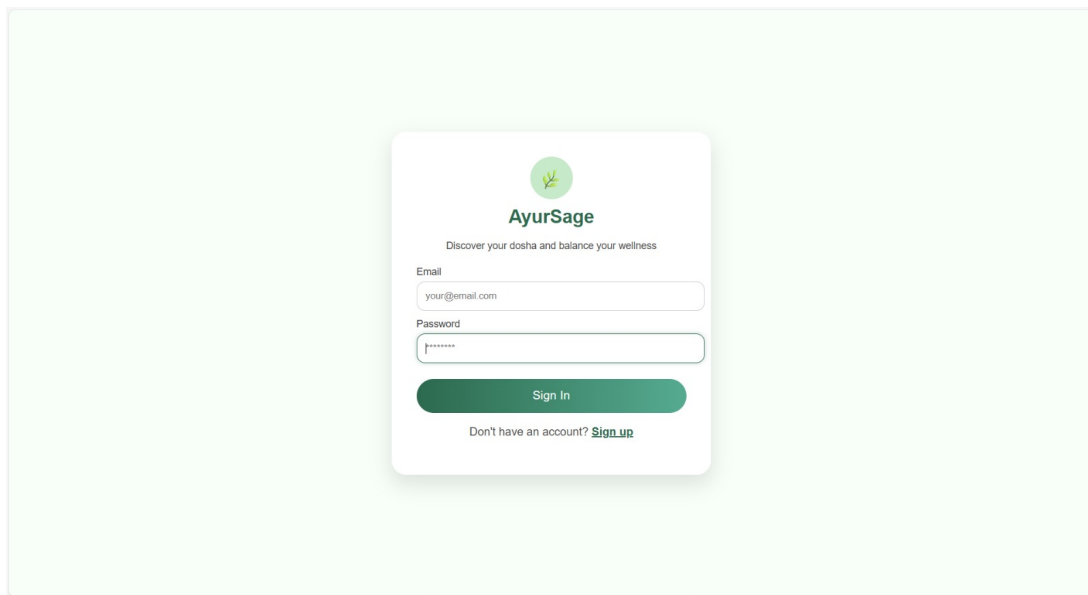
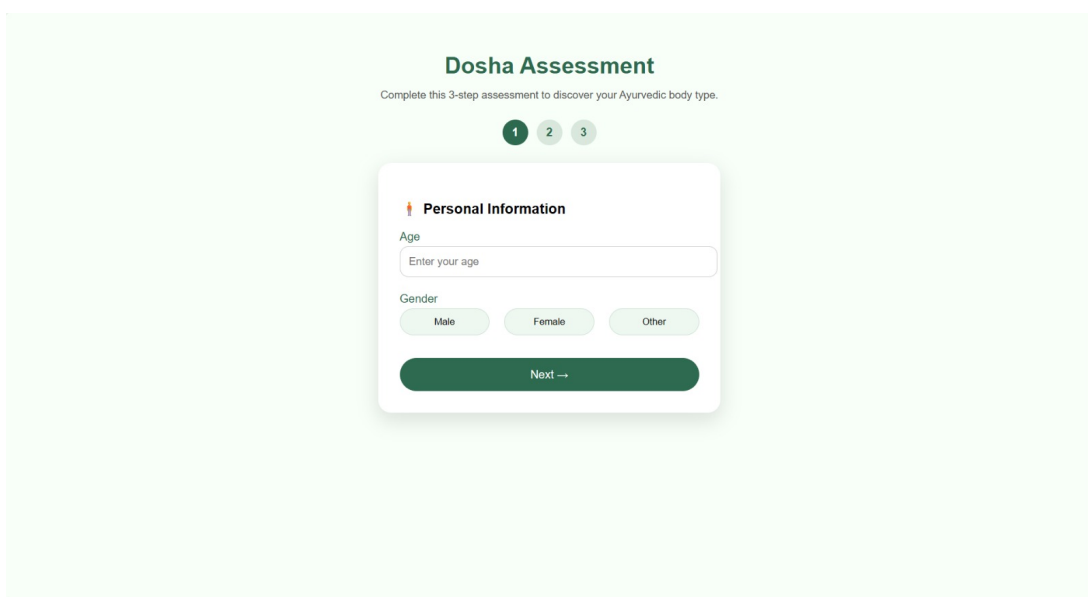


Figure 7: Dashboard



The login page for AyurSage features a central white card on a light green background. At the top of the card is a green circular logo with a white plant icon. Below the logo, the text 'AyurSage' is displayed in a bold, dark green font, followed by the tagline 'Discover your dosha and balance your wellness' in a smaller, lighter green font. The login form includes an 'Email' field with the placeholder 'your@email.com' and a 'Password' field with a masked input. A prominent green 'Sign In' button is positioned below the fields. At the bottom of the card, a link for users without an account is provided: 'Don't have an account? [Sign up](#)'.

Figure 8: Login Page



The 'Dosh Assessment' section is presented on a light green background. The title 'Dosh Assessment' is centered at the top in a bold, dark green font, with the instruction 'Complete this 3-step assessment to discover your Ayurvedic body type.' below it. A progress indicator shows three steps, with the first step (1) highlighted in green. The main form is a white card titled 'Personal Information' with a small flame icon. It contains an 'Age' field with the placeholder 'Enter your age' and a 'Gender' section with three radio button options: 'Male', 'Female', and 'Other'. A green 'Next ->' button is located at the bottom of the card.

Figure 9: Dosh Prediction Section1

Dosha Assessment

Complete this 3-step assessment to discover your Ayurvedic body type.

1
2
3

💡

Current Symptoms

Describe physical or emotional symptoms you're experiencing.

Example: Dry skin, anxiety, constipation...

← Previous
Generate Result 🌟

Figure 10: Dosh Prediction Section3

CONCLUSIONS, LIMITATION AND FUTURE WORKS

7.1 Conclusion

The AyurSage system has been effective in showing how Ayurvedic science can be applied to modern technology to develop a smart health advisory system. The system can combine rule-based logic with machine learning to diagnose the symptoms provided by the user and suggest potential Dosha imbalances and related health problems. The recommendations that are generated such as herbal remedies, dietary recommendations, and lifestyle changes offer the users a comprehensive system of enhancing their health.

Besides, the project emphasizes the opportunities of digital health solutions to make ancient medical practices more accessible and easier to use. The system will be interactive and informative in the sense that it will assist the users to obtain awareness concerning their health conditions in simplified form. Altogether, AyurSage can be regarded as a good basis of further developments of AI-based personalized healthcare systems in the field of alternative medicine.

7.2 Limitations

Even though AyurSage system is effective, it has some limitations which must be recognized. The use of a pre-existing knowledge base and logic based on rules is one of the most significant restrictions as it might not address all possible symptoms or diseases or personal variations. Ayurveda is very individualized and the system might not be able to capture all the complexity of the individual Prakriti (body constitution) and environment influence on health.

Also, the machine learning model in the system is trained on small or artificial datasets, which can affect the accuracy and extrapolation of predictions. The system does not substitute professional medical advice and might not be applicable in diagnosing serious or critical health conditions. Thus, users are to regard the results as hinting and not conclusive.

7.3 Future Work

The future of AyurSage is enormous and it can be expanded considerably through further development. The addition of larger, more realistic Ayurvedic datasets, such as clinical data and resources corroborated by experts, to increase the quality and reliability of predictions is one of the possible improvements. It is also possible to upgrade the system with the help of advanced machine

learning or deep learning models to learn more about complicated symptom patterns and become more individualized. Moreover, the capabilities of the system in the future can be enhanced with the help of real-time health tracking, support of mobile applications, multilingual interfaces, and wearable devices. The system can also be enhanced by including user feedback, consultation support by certified Ayurvedic practitioners which will help in making this system more practical and trustworthy. These improvements will assist in making AyurSage a more holistic and universal digital healthcare tool.

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